

REVIEW **OPEN ACCESS**

Comprehensive Review on Concentrated Solar Photovoltaics: Manufacturing, Cooling Technologies, and Advanced Applications

Abdul Ghani Olabi¹  | Adnan Alashkar¹  | Montaser Mahmoud¹  | Ayman Mdallal²  | Malek Kamal Rabaia¹  | Abdul Hai Alami¹ | Mahmoud Elgendi^{3,4}  | Mohammad Ali Abdelkareem^{1,5} 

¹Sustainable Energy & Power Systems Research Centre, Research Institute of Sciences and Engineering, University of Sharjah, Sharjah, UAE |

²Department of Civil Engineering, McMaster University, Hamilton, Canada | ³Department of Mechanical and Aerospace Engineering, College of Engineering, United Arab Emirates University, Al Ain, UAE | ⁴National Water and Energy Center, United Arab Emirates University, Al Ain, UAE |

⁵Chemical Engineering Department, Faculty of Engineering, Minia University, Al Minya, Egypt

Correspondence: Mahmoud Elgendi (mahgendi@uaeu.ac.ae)

Received: 24 September 2025 | **Revised:** 24 December 2025 | **Accepted:** 4 January 2026

Keywords: concentrated photovoltaic | concentrated photovoltaic applications | concentrated photovoltaic cooling | solar energy

ABSTRACT

Concentrated photovoltaic (CPV) systems offer a promising approach to enhancing solar energy conversion efficiency by utilizing optical concentrators and advanced solar cell technologies. This paper provides a comprehensive overview of the history, evolution, and fundamental characteristics of CPV systems. The paper also explores the strengths and limitations of CPV technologies, with particular attention to the materials used in the optical and solar cell components, as well as the manufacturing challenges that affect scalability and performance. A detailed analysis of cooling techniques, such as radiative, phase change material, liquid immersion, microchannel, and jet impingement, is presented to address thermal management in high-concentration environments. Accordingly, the differences among these cooling techniques in terms of energy consumption, reliability, and adaptability have been investigated. Furthermore, the paper examines the integration of CPV systems into advanced applications, including solar-powered desalination, thermoelectric generators, light-splitting configurations, and building-integrated structures. Moreover, digital integration in CPVs has recently been considered a promising approach to maximizing the electrical output of these systems, mainly through the enhancement of solar tracking systems and input/output predictions. Through this multi-dimensional review, the study highlights the potential of CPV technologies to contribute significantly to sustainable energy solutions, while also addressing the technical and practical challenges that remain.

1 | Introduction

On a day with cloudless skies, the maximum normal surface irradiance received by Earth at sea level is $\approx 1000 \text{ W/m}^2$, which represents only a portion of the solar radiation received, estimated at 1366 W/m^2 [1]. This establishes the sun as a significant and dependable source of energy production, giving it a huge advantage over traditional sources, mainly through environmental and technological viability [2]. When compared to their renewable counterparts, solar power systems are in an advantageous position

technologically, economically, and environmentally. This is attributed to the fact that a variety of solar systems and technologies have been commercialized and used in various fields on both large and small scales. Photovoltaics (PV) is the technology that allows for the direct production of direct current (DC) from a portion of the absorbed solar light [3]. From powering satellites since the beginning of spaceflight to charging batteries in communication systems, these solar cells have a multitude of applications.

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2026 The Author(s). *Advanced Energy and Sustainability Research* published by Wiley-VCH GmbH.

PV systems use a variety of concentrator types, such as parabolic solar concentrators [4], Fresnel lenses [5], compound parabolic concentrators (CPC) [6], crossed CPC [7], rotationally asymmetrical CPC [8], double receiver CPC [9], and dielectric-filled CPC [10]. While PV and concentrated solar power (CSP) technologies have had the lion's share in attention and over empowerment of other solar and nonsolar technologies owing to their outstanding compatibility with utility-scale power production and easier installation requirements, the continuously developing and incredibly high efficiencies of the new technology of concentrated photovoltaic (CPV) are rather promising and worth investment. The primary concept that differentiates CPV technologies from other solar technologies is the utilization of cost-efficient optics to focus and concentrate sunlight on solar cells. In turn, this allows for a significant reduction in the receiving area of the solar cell, depending on the concentration ratio (CR), with an increase in light intensity [11]. Such technological characteristics, in most designs, if not all of them, require a continuous direct irradiance. CPV systems typically utilize multijunction solar cells (MJSCs) due to their superior ability to harness a broader range of the solar spectrum under high irradiance conditions. By focusing sunlight onto a small, high-efficiency cell area using cost-effective optical components, CPV systems significantly reduce the overall material cost and minimize energy losses associated with larger surface areas. These systems employ diverse optical assemblies and configurations to direct sunlight from a broad, low-cost collection area onto a compact, high-cost photovoltaic cell. A critical performance metric in CPV technology is the CR, which quantifies the quantity of solar irradiance received by the PV cell relative to the aperture area of the concentrator. CPV systems are typically classified into three categories based on the CR ratio; for instance, CPVs with a CR ratio less than 40 are dubbed as **low concentration PV (LCPV)**. Medium concentration PV (MCPV) usually refers to systems with CRs between 40 and 300, and **high concentration PV (HCPV)** systems utilize CRs higher than 300 [12].

MJSCs exhibit the highest efficiency among all PV technologies. This superior performance stems from their multiple semiconductor

layers, which enable broader solar spectrum absorption and, consequently, greater electrical conversion efficiency [13]. In comparison, conventional multicrystalline silicon solar cells achieve a maximum efficiency of $\approx 23.3\%$, as shown in Figure 1 obtained from the National Laboratory Energy Research (NREL) [14]. By contrast, six-junction solar cells developed by the Fraunhofer Institute for Solar Energy Systems have achieved record efficiencies as high as 47.6% [15]. Despite these impressive figures, a significant fraction of the absorbed solar energy is inevitably converted into heat, leading to thermal losses and reduced electrical output. While MJSCs and minimized active areas can lower the cost per watt in CPV systems, the overall system costs are elevated due to the need for advanced concentrator optics and precise solar tracking mechanisms, often requiring single- or dual-axis tracking for optimal performance. Moreover, because CPV systems operate under high solar CRs, they are subject to elevated operating temperatures, necessitating robust thermal management solutions. Although effective, these cooling strategies, ranging from passive techniques to active systems, introduce additional design complexity and expense, limiting the widespread commercial adoption of CPV technology. Nonetheless, the thermal energy generated in CPV systems presents valuable opportunities for cogeneration applications. This waste heat can be repurposed for powering thermoelectric generators (TEGs), accelerating drying processes, driving solar-assisted desalination systems, and supporting passive building ventilation and heating. Such integrated energy utilization strategies can enhance the overall efficiency and economic viability of CPV systems in specialized markets.

This paper aims to deliver an in-depth and multidisciplinary evaluation of CPV systems, with a focus on historical development, core characteristics, and technological advancements. The study dissects the materials and manufacturing processes used in CPV, including optics and solar cell components, while critically addressing the scalability and fabrication challenges that hinder commercial adoption. A central aspect of the study is the comprehensive comparison of advanced cooling strategies, such as radiative, phase change material (PCM), liquid immersion,

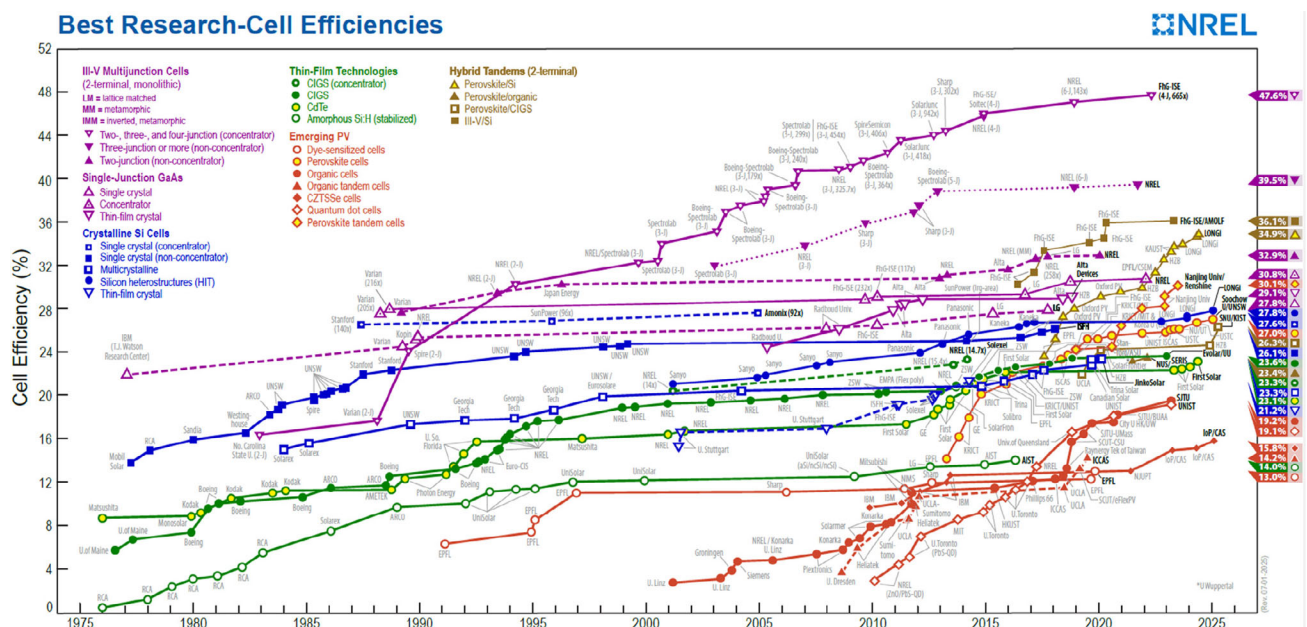


FIGURE 1 | Best solar cell efficiencies achieved [14] (Open Access, obtained from NREL).

microchannel, and jet impingement cooling, highlighting their effectiveness in mitigating thermal stresses under high solar concentration. In addition, the paper explores the emerging applications of CPV in solar desalination, hybrid thermoelectric systems, light splitting, and building-integrated photovoltaics, offering insight into their potential to transform both energy and water infrastructure. A key novelty of this work lies in its integration of conventional CPV analysis with modern innovations, such as the inclusion of digitalization technologies, such as artificial intelligence (AI), machine learning (ML), and Internet of Things (IoT).

2 | History of Concentrators and CPV

In the past, concentrating optics were used to their full potential in the creation of heat for a variety of applications. For instance, certain Roman documents claimed that Archimedes employed concentrating mirrors to set fire to Roman ships by focusing the light of the sun [16]. In the 19th century, innovators and engineers used heat created by sun concentrators to power water-pumping steam engines [17]. Furthermore, in the 1970s and after almost two decades of developing solar cells, engineers started demonstrating the idea of enhancing the solar PV's efficiency and possibly decreasing its cost through the concentration of several hundred suns on the receivers of said cells. The first effective and officially recognized creation of a CPV technology occurred in 1976, when the Sandia National Laboratories in the United States developed a total of 1 kWp array of CPV employing Fresnel lenses and two-axis tracking systems operated by an analogue closed-loop tracking system with passive cooling (Sandia I & II prototypes) [18]. Figure 2 depicts the first modern SANDIA II array that was built in New Mexico in the year 1977 and was designed to function under a concentration of 32 suns, relying on passive cooling. Building on the work, Sinton et al. [19] produced solar cells with point contact concentrators using a variety of designs based on different materials, such as silicon (Si), gallium arsenide (GaAs), and so on. These cells showed an efficiency of 23–28% and were able to survive high CRs. However, given the high capital cost required and low manufacturing yield, the technology never flourished.



FIGURE 2 | The first SANDIA II CPV array by Sandia National Laboratories in 1977 in New Mexico [18] (With Permission, License Number: 6 114 041 481 089).

Conventional concentrator cells required complex manufacturing processes, specialized optical components, and continuous solar tracking, which limited their scalability and production capacity. To address these challenges, researchers proposed using bifacial solar cells as a substitute for specialized concentrator cells and their costly tracking systems. Bifacial solar technology builds on the concept of static light concentration proposed by Hinterberger and Winston for Cerenkov radiation [20]. Thus, a new direction began to emerge as an alternative to traditional concentrator photovoltaic systems. This approach aims to offer a modular, static solution—similar to conventional flat-plate panels—with no moving parts, yet capable of efficiently collecting and concentrating diffused solar radiation. The envisioned system could maintain a nominal voltage of 12V while operating without active tracking mechanisms. Development efforts for this concept continued into the early 2000s. However, despite requiring only modest additional investment, the technology was never commercialized. The main barriers were the perceived risk and market uncertainty associated with introducing an unfamiliar technology, and the insufficient profit margins to justify the investment relative to the effort required [21].

More recent technologies were developed employing GaAs- and Si-based cells contained inside angular confining cavities. These technologies reached high efficiencies up to 29.4% for GaAs cells and 35.4% for Si cells at elevated CRs of 666 and 676, respectively [22]. One of the most significant efforts in CPV industrialization during the 1990s was led by ENTECH in Texas. The company developed multiple 100-kWp systems featuring dual-axis tracking and curved Fresnel lenses reaching a CR of 20 suns. These systems achieved an optical efficiency of 85%, marking a major milestone in the commercialization of CPV technology during that decade [23]. Recently, advancements in CPV have been focused on III–V MJSCs due to their extraordinarily high efficiencies surpassing 45% [24]. Given that the MJSCs are theoretically limited to an efficiency of 86%, which is more than twice that of single-junction cells (around 40%), there has been significant motivation to pursue their development. While single-junction cells are primarily constrained by the Shockley–Queisser limit, multijunction designs overcome this by absorbing a broader portion of the solar spectrum. These compelling theoretical advantages, supported by promising experimental results, have prompted the exploration of the manufacturing of such technologies, particularly in conjunction with concentrator systems to further enhance their performance [25]. Furthermore, MJSCs are often used for extraterrestrial applications where their high cost is justified by performance needs. This economic constraint is a key reason why the HCPV sector has focused on commercializing triple-junction and MJSCs [26]. These advanced cells have achieved remarkable and continuous improvements in efficiency, driven by enhancements in several critical areas: wafer fabrication techniques, epitaxial layer quality, and innovative cell designs [27]. The vast majority of commercially available MJSCs contain lattice-matched semiconducting layers deposited on germanium (Ge) substrates by chemical processes such as vapor deposition [28]. Table 1 summarizes the most notable CPV systems utilizing III–V MJSCs, with verified efficiencies exceeding 41%, as reported by recognized testing laboratories.

Perovskite solar cells have received extensive attention in the past years due to their facile and low-cost fabrication, yet high efficiency that reaches around 27% [14]. In addition, perovskite solar

TABLE 1 | Most remarkable (more than 41% efficiency) CPV technologies, in particular III and V MJSCs.

Ref.	Institution	Cell design	Efficiency %	Comments
[29]	Fraunhofer ISE/ Soitec/ CEA	Quadruple-Junction GaInP/GaAs/GaInAsP/ GaInAs	46.0% at 508CR	Lattice-matched
[30]	NREL	Quadruple-Junction GaInP/GaAs/GaInAs/ GaInAs	45.7% at 234CR	Inverted metamorphic
[31]	Sharp	Triple-Junction GaInP/GaAs/GaInAs	43.5% at 306CR	Inverted metamorphic
[32]	Solar junction	Triple-Junction GaInP/GaAs/GaInNAs	43.5% at 925CR	Lattice-matched
[33]	NREL	Triple-Junction GaInP/Ga(In)As/GaInAs	42.6% at 327CR & 40.9% at 1093CR	Inverted metamorphic
[34]	Spire	Triple-Junction GaInP-GaAs-wafer-GaInAs	42.3% at 406CR	On the front of GaAs wafer: epi growth lattice-matched
[35]	Spectrolab	Triple-Junction GaInP-Ga(In)As-Ge	41.6% at 364CR	Commercially available, Lattice-matched
[36]	Fraunhofer ISE	Triple-Junction GaInP-GaInAs-Ge	41.1% at 454CR	Commercially available and upright metamorphic

cells possess tunable opto-electronic properties that make them a great candidate for tandem devices. For instance, perovskite/silicon tandem solar cells achieved a power conversion efficiency over 30% [14]. Although these efficiencies are already high, the growing demand for solar energy necessitates further efficiency gains. One promising route is the integration of perovskite solar cells with CPV technology. Recent studies have demonstrated significant performance enhancement in perovskite/CPV tandem systems, achieving a peak efficiency of 34.62% under 70 suns, compared with 30.52% under one-sun illumination [37].

3 | Characteristics and Manufacturing of CPV

HCPV and LCPV are the two broad categories of CPV technology, which utilize different tracking systems and consist of various solar cell types, as shown in Table 2. Systems operating at concentrations between 40 and 300 are typically classified as medium-concentration photovoltaic. However, due to their lack of competitiveness in terms of the levelized cost of electricity (LCOE), they have seen limited adoption in commercial applications.

TABLE 2 | Classification of CPV technologies.

Class	CR	Tracking system	Cooling	Type
HCPV	Above 300	Two-axis	Active	III–V MJSCs
MCPV	40–300	Two-axis	Active or Passive	Mostly III–V MJSCs
LCPV	Below 40	One- or Two-axis	Passive	Crystalline Si

3.1 | Strengths and Weaknesses of CPV

Since CPV integrates elements of both CSP and PV systems, it possesses a unique set of advantages and limitations compared to other solar technologies. CPV systems focus sunlight, often by a factor of tens, hundreds, or even thousands, on specially designed PV cells, converting the concentrated light into electricity. The benefits and challenges of CPV technology are summarized in Table 3. As with most emerging technologies, CPV has exhibited a balance of strengths and weaknesses, particularly considering that it only began to enter the commercial solar market in the mid-2000s. However, it is noteworthy that the relative influence of each factor matters more than the number of advantages or disadvantages. In some cases, a minor drawback may have little economic impact, while a single strength—such as a significant gain in efficiency—can be transformative for system performance and competitiveness.

3.2 | Materials and Manufacturing of CPV

The optics and solar cells are the two most critical components in any CPV system. While modern CPV systems typically utilize III–V MJSCs, originally developed for strenuous applications such as aerospace, the optical design is equally essential and must be tailored specifically for CPV requirements. This includes careful selection of materials, geometries, and configurations to optimize light concentration and system efficiency.

In designing effective CPV optics, two key factors must be balanced: achieving high CRs and maintaining tight optical tolerances, all while minimizing material usage and reducing the area required for expensive high-efficiency solar cells. To guide this balance, the concentration–acceptance product (CAP) factor is introduced as a performance metric. It quantifies the trade-off between concentration efficiency and angular acceptance, enabling more informed evaluation and optimization of optical designs.

TABLE 3 | List of CPV's strengths and weaknesses (Adapted from [38]).

Strengths	Weaknesses
- Exceptionally high theoretical efficiency (~86%) compared to single-junction PV (~40%), enabling better land use and reduced system costs. - Low temperature coefficients ensure stable performance in high-heat environments. - No water is needed for passive cooling systems, unlike CSP technologies. - Potential for waste heat recovery through integration with active cooling systems. - Highly modular and scalable—from kilowatt to gigawatt scales. - Dual-axis tracking enables consistent and increasing energy yield throughout the day. - Short energy payback time, improving return on investment - Low environmental impact, with potential for dual land use (e.g., agri-voltaics). - Opportunity for local manufacturing of components, enhancing regional economic benefits. - Reduced sensitivity to semiconductor material prices, thanks to a small active cell area.	- Limited ability to utilize diffuse sunlight, particularly in high-concentration CPV systems. - Requires precise and reliable tracking systems, increasing system complexity and cost. - Susceptible to soiling; frequent cleaning may be needed in dusty or arid locations. - Limited suitability for rooftop or urban settings; effectiveness is constrained to regions with high normal irradiance. - Slower cost reduction trends compared to mainstream PV technologies. - Perception and Bankability problems due to historical differences compared to PV - Market adoption is hampered by bankability and investor perception issues due to a lack of commercial history. - Perceived technological risk due to limited deployment and production history. - Optical losses from reflection, absorption, and transmission within concentrator optics. - Lack of standardized designs and technical protocols limits large-scale adoption.

Before improving or comparing existing systems, it is essential to clearly define and analyze such performance variables.

$$CAP = \sqrt{C_g} \sin \alpha \quad (1)$$

where α is the acceptance angle, and C_g is the geometric concentration. The acceptance angle typically refers to the incidence angle where 90% of the on-axis power is collected by the concentrator. The CAP is generally capped at 1.42, which is the refractive index of Si, the medium that surrounds the cell [39]. This limit is consequent of the conservation of etendue, a principle that governs the preservation of light flux in optical systems. As such, the CAP serves as a key metric for determining the maximum light concentration capability of an optical design relative to its angular acceptance range [40, 41]. In addition, the optical efficiency of a CPV system refers to the fraction of incident solar power successfully transmitted through the optical elements and delivered to the cell. This efficiency is wavelength-dependent and is typically used by optical component manufacturers to optimize materials and coatings that minimize spectral mismatch with the photovoltaic cell's absorption characteristics. Optical efficiency is often evaluated by spectral weighting using the ASTM G173 Air Mass 1.5D solar spectrum standard.

3.2.1 | Optics' Materials in CPV

The primary optical elements (POE) and the secondary optical elements (SOE) are the backbone of CPV technology's optical design. The primary function of the POE is to concentrate

sunlight onto a small section of the solar cell, whereas the primary function of the SOE is to divert and evenly disperse sunlight over the solar cell [42, 43]. When put together, these factors improve the CPV system's optical efficiency, making it less sensitive to tracking errors and preventing the solar cells from being exposed to heterogeneous spatial and spectral light [44, 45].

POEs may be constructed from a variety of materials, including mirrors, polymethylmethacrylate (PMMA), and silicone-on-glass (SoG) Fresnel lenses. Mirrors are commonly employed in LCPV systems with Si-cells, although they are seldom utilized as POEs nowadays. Mirrors are composed of glass or aluminum-coated silver to increase reflectivity over the entirety of the electromagnetic spectrum. Fresnel lenses constructed from PMMA and SoG are viable solutions. SoG often offers superior performance and is the industry standard [46]. The primary differences between PMMA and SoG Fresnel lenses' structures are summarized in Table 4, and more nuanced distinctions can be found in the comprehensive study by Miller et al. [47].

SOEs are often made of render glass, whereas reflective SOEs are typically made of silver or aluminum. Molded components usually consist of Glass type B270, while flat components and surfaces are made up of glass type BK7. A new molded glass competitor made from a colloidal suspension of silica particles has been developed by Evonik. This material outperforms glass in many ways, including its ability to withstand ultraviolet light, its high transmission efficiency across the entire solar spectrum, its superior response to thermal expansion, and its flexibility [48].

In addition to their structural and optical differences, PMMA and SoG Fresnel lenses also exhibit distinct long-term performance,

TABLE 4 | Major differences between main POE options: PMMA and SoG Fresnel lenses [47].

Fresnel lenses	
PMMA	SoG
- Based on PMMA. - Sensitive to oxidative photodegradation. - Lighter weight. - Easy manufacturing processes. - No potential risk of glass solarization or corrosion. - Lower possibility of thermal expansion negatively affecting optical densities and refractive indexes.	- Based on silicone Fresnel facets and glass cover. - Glass offers protection, stabilizes and supports the structure. - Although being based on two components, there are no Fresnel losses due to them having the same refractive index values. - Considerably higher market share. - Higher transmission efficiency. - Broader spectrum and better for spectrum splitting systems. - Better resistance to external effects like scratches and chemicals.

cost, and application characteristics that are critical for practical CPV deployment. PMMA lenses generally offer lower upfront manufacturing costs due to simpler molding and fabrication processes, enabling cost-effective production; however, their long-term optical durability in outdoor conditions is a concern. Several studies have reported that PMMA Fresnel lenses can experience discoloration, loss of transmittance, hazing, mechanical fatigue, and warping over extended exposure to solar irradiance, which may degrade performance over time if not adequately protected or replaced at regular intervals [47, 49, 50]. In contrast, glass lenses with a rigid glass substrate exhibit better resistance to environmental aging, scratching, and abrasive wear, with experimental evidence suggesting that glass-based optics retain transmittance and conversion efficiency significantly longer than PMMA under harsh conditions such as desert sand exposure. For instance, according to the study conducted by Nishioka et al. [51], it was found that the durability of glass Fresnel lenses is approximately twice that of PMMA lenses under equivalent sandblasting conditions. While SoG manufacturing may involve higher initial material and processing costs relative to polymer molding, the longer service life and reduced maintenance or replacement requirements can lead to lower lifecycle costs in demanding CPV applications. Therefore, PMMA Fresnel lenses are typically suited for lower-cost, medium-lifetime deployments, or regions with less severe environmental stress; whereas SoG lenses are often preferred for high-reliability and long-lifetime CPV installations where optical stability and minimal degradation are essential for long-term applications [52].

3.2.2 | Solar Cell Materials in CPV

The production of modern III–V MJSCs primarily depends on the critical elements gallium (Ga), indium (In), and Ge. According to reports published in December 2015 by NREL and Fraunhofer ISE on the status of CPV technology, the estimated annual production of these elements was $\approx 280, 770$, and 118 metric tons, respectively [38]. These figures represent the output of “virgin” materials only, excluding any contributions from recycling processes.

A comprehensive resource availability assessment conducted by Frenzel et al. [53] highlighted potential long-term constraints on the supply of these critical materials. As the demand for longer-lifetime systems grows, the low recycling rates of Ga, In, and Ge may pose a significant challenge. If the required material input for manufacturing exceeds the primary production rate, it could hinder the scalability and sustainability of future CPV deployment.

3.2.3 | Challenges in Manufacturing CPV

The primary challenges associated with CPV technology are predominantly linked to HCPV, due to the demanding operational conditions required to achieve consistent and efficient performance. One notable example is the widespread use of flat Fresnel lenses as concentrators in CPV systems. These image-forming optics are designed to concentrate solar radiation onto the solar cell, but they must produce a solar image large enough to accommodate alignment tolerances, such as the acceptance angle. However, the inherent discontinuities in the nadir magnification of these lenses often result in a blurred solar image with a bell-shaped intensity distribution, rather than the ideal stepped profile. Furthermore, thermal management and performance stability can be hindered by the structural configuration of the solar cell or the module itself. For instance, positioning a GaInP/GaAs solar cell in a downward orientation within the module can complicate the design of an effective heat sink. Additional challenges include the logistical and operational costs related to transportation, system monitoring, and maintenance. A promising strategy to address these issues involves minimizing the overall weight and size of the system, thereby simplifying deployment and improving system reliability.

Wiesenfarth et al. [54] provided a comprehensive overview of the key challenges that CPV technologies must address to achieve higher performance and efficiency. A significant portion of these challenges arises from the disparity between ideal design and prototyping conditions and the complex realities of actual operational environments [55]. For example, optical concentrators rarely deliver uniform solar illumination across the surface of the solar cell. Thus, the reported concentration level often represents an average of a nonuniform concentration profile. Accurately characterizing this illumination profile is critical for optimal solar cell matching, and numerous research groups have shared their findings and methodologies for dealing with such conditions.

This nonuniform light distribution leads to thermal gradients within the cell: a horizontal gradient caused by spatial variation in light intensity, and a vertical gradient resulting from differences in absorption across the cell’s semiconductor layers. Understanding the full 3D temperature distribution, driven by the illumination profile, is essential for modeling, predicting, and mitigating localized heating effects and hot spot formation [56]. Such insights are crucial for enhancing thermal management strategies and ensuring long-term cell reliability.

4 | Cooling Techniques for CPV

To achieve and maintain the high performance and efficiency level of CPV, cooling technologies are essential. Compared to conventional PV systems, CPV systems provide higher efficiency because of the utilization of lenses or mirrors to direct and focus solar radiation onto specific regions of solar cells. Thus, an increase in the operating temperature is expected, and in turn, a drop in efficiency is anticipated. CPVs are frequently designed to operate in areas that receive intense daily sunlight, and their temperatures frequently rise above 200°C; hence, cooling becomes a predominant need for the continuous function of these systems [57]. Various cooling technologies are available, including radiative cooling, the utilization of liquid or PCMs, and microchannel heat sinks.

4.1 | Radiative Cooling

Radiative cooling is based on the emission of infrared radiation from the surface of the CPV module to the cold sky, thereby dissipating excess heat without the need for active components like fans or fluid loops. Thus, it can be labeled as a passive cooling mechanism. Effective radiative cooling can help maintain the temperature of the CPVs closer to their optimal operating range, improving energy conversion efficiency and reducing the risk of thermal degradation. Integrating radiative cooling strategies into CPV designs is essential for optimizing thermal management, particularly in remote or arid regions where conventional cooling methods may be impractical or energy-intensive. In both computational and experimental research, Wang et al. [58] employed various radiative cooling techniques to cool CPV cells. The experimental results showed that by employing a heat load of 6.1 W, the operating temperature in a passive cooled CPV can be lowered by 36°C. As a result, the open-circuit voltage (V_{oc}) has increased by a relative 27%, and the anticipated CPV lifespan increased by 4–15 times. Figure 3 shows the experimental system setup, and the temperature and open circuit voltage variation with time.

Nishioka et al. [59] applied a thermal radiation layer on the metal chassis of a CPV module. The presence of the layer decreased the

temperature of the CPV roughly 10°C, dropping it from 93.1°C to 82.6°C. This improvement suggests that the thermal radiation coating enhanced heat dissipation by serving a dual function—facilitating both conductive and radiative heat transfer. Zhou et al. [60] fabricated a radiative cooler composed of a multilayer stack of composite materials, achieving a temperature reduction of 10°C. This cooling effect led to a 5.7% increase in V_{oc} and an estimated 40% increase in the operational lifespan. Using a predictive model, it was predicted that under 35 suns of concentration, the same design could yield a 34% increase in V_{oc} . Consequently, for failure mechanisms with high activation energies, the LCOE could be reduced by up to 33%.

Sun et al. [61] investigated the impact of radiative cooling on HCPV systems by designing a cooling structure consisting of a diamond heat spreader layer and low-iron soda-lime glass coated with a porous antireflection layer. This configuration exhibited high mid-infrared emissivity while maintaining strong solar transmittance, enabling effective radiative cooling under direct sunlight with minimal loss of concentrated solar energy. The proposed radiative cooler could achieve temperatures up to 14 K lower than those obtained with a conventional copper cooler.

Li et al. [62] proposed a heat dissipation idea that combines a finned heat sink with a radiative cooling layer (RCL). By positioning the RCL above the heat sink, the CPV's ability to dissipate heat was improved, and thermal radiation effects were enhanced. The RCL's contributions to lowering device temperature, improving heat dissipation, and reaching a manageable CR under various external conditions were examined. In addition, the influence of ambient temperature and wind velocity on the RCL's cooling efficiency was investigated. Numerical simulation results showed that even under extreme conditions (wind velocity of 0 m per second; ambient temperature of 50°C), the proposed passive cooling system can guarantee that the solar cell's temperature stays below 71.5°C at a CR of 200. Additionally, the device can ensure that the greatest temperature differential within it is less than 3.8°C. This agrees with the requirements for heat dissipation in silicon solar cells with low-medium concentration. Figure 4 shows the proposed CPV heat dissipating device, in which the concentrated

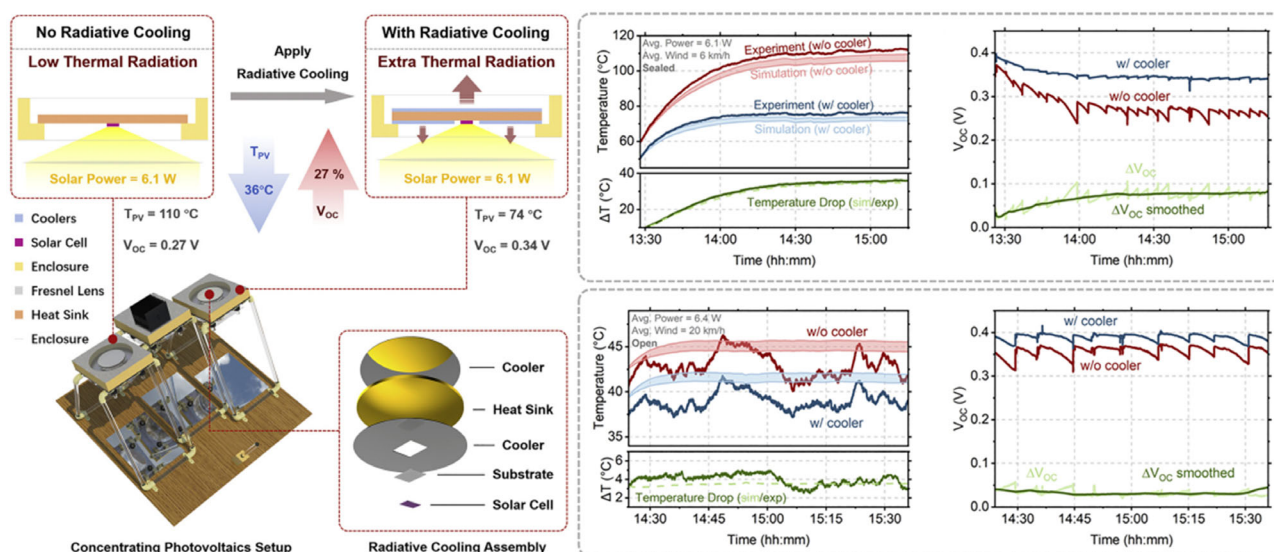


FIGURE 3 | A radiative experimental setup device, with experimental and simulation findings (temperature, and open circuit voltage) [58] (With Permission, License Number: 6 112 460 812 800).

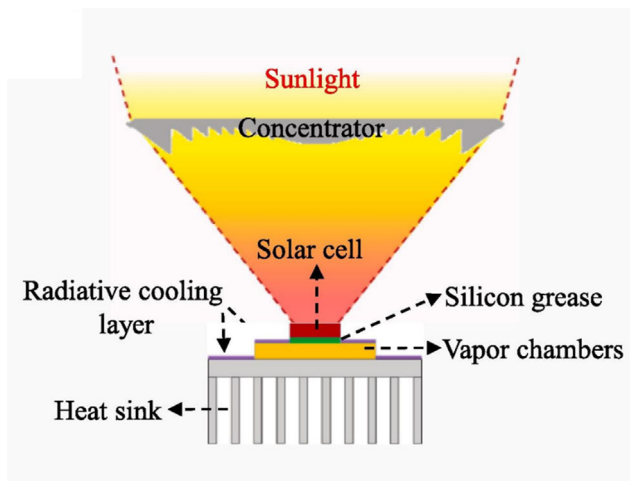


FIGURE 4 | Point-focusing and heat-dissipating CPV cooling device [62] (With Permission, License Number: 6 111 870 123 005).

sunlight is directed onto a silicon solar cell that is coupled to the vapor chambers through a thermal interface material, while the chambers are welded to a heat sink, except for the region occupied by the solar cell. Heat generated in the solar cell is transferred through the thermal interface material to the vapor chambers and subsequently to the heat sink, where it is dissipated to the surroundings via convection and radiation.

Wang [63] designed and fabricated an innovative structure aimed at increasing the concentration factor and improving the thermal management of CPV systems. Experiments were conducted using a GaSb solar cell under various environmental conditions, with the system concentrating ≈ 39 suns of solar energy. While GaSb cells can experience more pronounced efficiency losses compared to silicon cells under thermal stress and potentially limit overall efficiency, the integration of a double-sided soda-lime glass radiative cooler significantly mitigated this issue. The cooler enabled a steady-state temperature reduction of 36°C and an increase in V_{oc} of ≈ 0.8 V. This improvement is estimated to extend the solar cell's operational lifespan by a factor of 4–13.

4.2 | PCM Cooling

PCM cooling offers an effective passive thermal management solution for CPV systems, particularly under fluctuating solar intensities and high irradiance conditions. PCMs absorb excess heat during peak solar exposure by undergoing a phase transition, typically melting, thereby stabilizing the temperature of the cells and preventing thermal degradation. With the presence of temperature regulation, the CPV cells can function near their optimal operating point, improving power output and extending system lifespan. During periods of lower irradiance or nighttime, the stored heat is gradually released as the PCM solidifies, making the system inherently self-regulating. The integration of PCMs into CPV modules is especially valuable in remote or off-grid locations where active cooling systems are impractical or energy-intensive [64].

Aoul et al. [65] designed a CPV-PCM/thermal device by placing the PCM on the opaque side of the CPV. In order to assess the cooling system's feasibility for PV cooling and water heating, an experimental evaluation was conducted in the hot climate of the United Arab Emirates (UAE). The CPV-PCM/Thermal system

lowered the average temperature of the CPV modules by 28°C at peak times, which increased the electrical production throughout the summer. The cost for CPV-PCM was found to be $\$0.18/\text{kWh}$, whereas the cost for a flat plate collector was $\$0.25/\text{kWh}$, indicating that the CPC-PCM system can produce energy at 28% less cost. Emam and Ahmed [66] developed an innovative combination of CPV-PCM-water jacket that offers high energy storage density during the day and improves heat removal from the PCM during the night. The system consisted of a CPV cell with either a single, three, or five cavities for the PCM heat sink. Figure 5 shows the proposed system design. As shown in the figure, the CPV cell is mounted directly onto the front surface of the PCM heat sink, which is integrated with a wide-channel water jacket of uniform 6 mm height. This water jacket surrounds the PCM heat sink on the top, bottom, and rear sides. If a single cavity design with a CR of 20 is utilized, the temperature of the solar cell can be lowered significantly (around 124°C) compared to the normal CPV-PCM system. Furthermore, at the end of the night, more than half of the liquid phase had transitioned into the solid phase, while the normal CPV-PCM only observed $\sim 10\%$ of the same transition. During the day, the temperature can be maintained at or below 72°C by employing a five-cavity design.

Rabie et al. [67] enhanced the cooling for LCPV systems by introducing a novel heat sink design featuring extended aluminum fins positioned outside a PCM container. Figure 6 shows the LCPV-PCM setups. A comprehensive analysis comparing various heat sink configurations was conducted, evaluating parameters such as temperature, efficiency, and energy balance under different CRs. The investigated configurations were compared to a reference case, where the CPV cell was mounted on a 3 mm aluminum plate, representing the uncooled scenario. As shown in the figure, four different heat sink configurations were studied: (a) straight aluminum fins with a thickness of 2 mm, (b) PCM tank, (c) straight aluminum fins with PCM filling the spaces between them, and (d) similar to the previous configuration but with the fins extended outside the PCM tank. At a concentration of five suns, the incorporation of PCM significantly improved thermal regulation, maintaining the temperature at 60°C during the phase change process. In contrast, conventional designs using aluminum heat sinks filled with PCM allowed the cell temperature to rise to 120°C , and up to 125°C when using

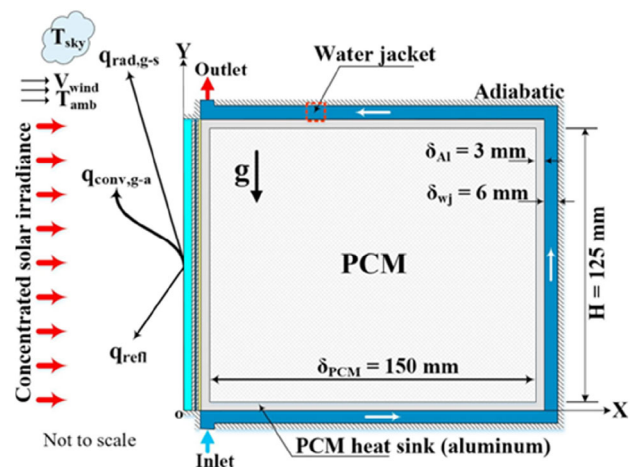


FIGURE 5 | CPV-PCM-water jacket system design [66] (With Permission, License Number: 6 112 461 257 569).

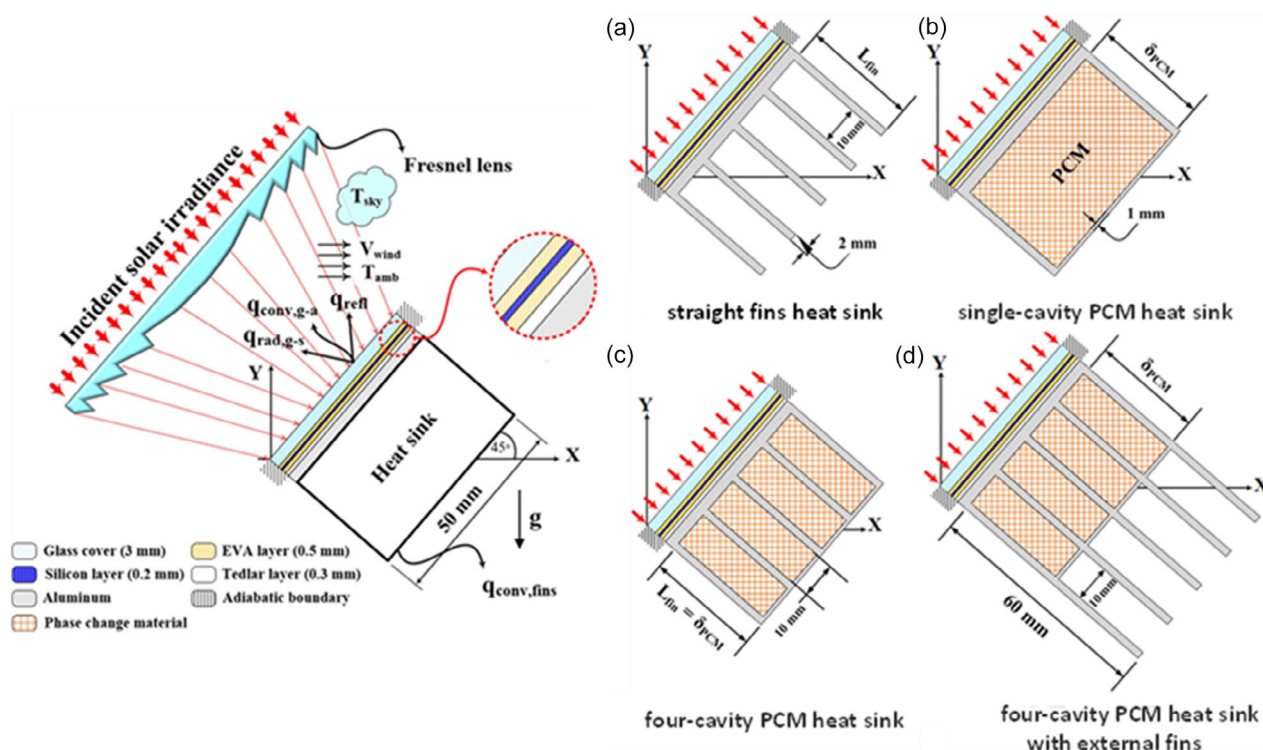


FIGURE 6 | Schematic diagrams of difference heat sink configurations for LCPV-PCM [67] (With Permission, License Number: 6 112 461 466 233).

four parallel cavities. The newly proposed design, however, achieved a temperature of just 82°C by extending the PCM melting duration and effectively delaying the temperature rise after the phase transition. This improved configuration not only stores thermal energy more efficiently but also maintains the temperature below the allowable limit, ensuring better performance and longevity.

4.3 | Liquid Immersed Cooling

Liquid-immersed cooling is an advanced thermal management technique that offers effective temperature control for CPV systems, especially under high solar concentration levels where conventional passive cooling may be insufficient. In this technique, the solar cells or the heat-dissipating components are submerged in a thermally conductive, electrically insulating liquid, such as mineral oil or engineered dielectric fluids, which absorbs and transports heat away from the active regions. This immersion allows for highly efficient convective heat transfer, maintaining lower and more uniform cell temperatures, thus enhancing the electrical performance and reliability of the CPV system. Additionally, liquid-immersed cooling can mitigate localized hot spots, reduce thermal stresses, and enable higher CRs without exceeding thermal limits. The effectiveness of this method depends on the thermophysical fluid properties such as thermal conductivity, specific heat capacity, and long-term chemical stability. While it introduces additional design and maintenance considerations, liquid-immersed cooling provides a promising solution for sustaining high efficiency and extending the operational lifespan of CPV modules in demanding environments [68].

Using a Fresnel-based CPV system, Hasan et al. [69] examined the impact of cooling MJSCs using water/ Al_2O_3 nanofluid. The utilization of the nanofluid greatly enhanced the transfer of

heat and lowered the cells' top surface temperature. Using Computational fluid dynamics (CFD) modeling, the performance of pure water and water/ Al_2O_3 nanofluid for cooling was compared in a small channel under concentrated MJSCs at Reynolds numbers ranging from 15 000 to 30 000. Nusselt numbers were found to be higher for higher Reynolds numbers, and a 23% improvement in heat transfer was achieved over pure water in a Fresnel-based CPV system. Employing water/ Al_2O_3 nanofluid led to a 21% improvement in the Nusselt number at a Reynolds number of 15 000 and a 23% improvement at a Reynolds number of 30 000. While employing water/ Al_2O_3 nanofluid as the cooling fluid, the top surface temperature dropped 10°C at the lowest Reynolds number of 15 000°C, and 8°C at the maximum Reynolds number of 30 000.

Sun et al. [70] demonstrated the effectiveness of direct liquid-immersion cooling using dimethyl silicone oil for linear CPV systems. In the experimental setup shown in Figure 7, a narrow rectangular channel receiver was tested under an energy flux of 9.1 suns, showing strong heat transfer performance. As illustrated in the figure, the receiver consisted of a series-connected array of 10 cells, a glass cover, a rubber gasket, and an aluminum substrate. The glass cover, gasket, and substrate were fastened together with bolts, creating a narrow rectangular channel in which the solar cell array was centrally positioned. Two bulges at the bottom served as supports for the cell tabs. According to their findings, monocrystalline silicon solar cells maintained stable conversion efficiency over 270 days of immersion under real-world environmental conditions. A linear relationship was observed between the temperatures of the oil and the solar cells, with typical cell temperatures ranging from 20°C to 31°C and a temperature difference of 5°C to 16°C between the cell and the cooling fluid. These results were achieved under a direct normal irradiance (DNI) of 910 W/m² and Reynolds numbers ranging from 2,720 to 13,602. The system

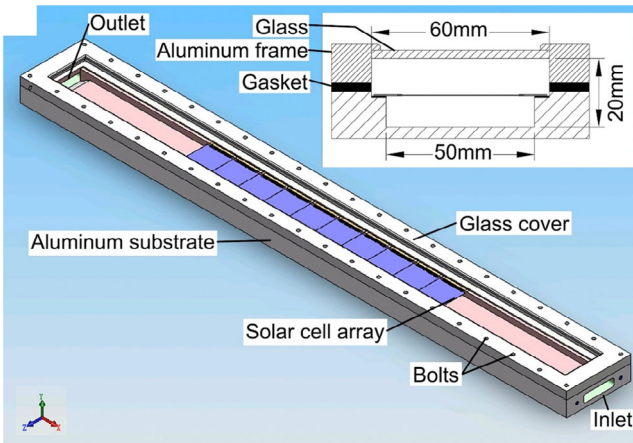


FIGURE 7 | A schematic and cross-sectional depiction of a narrow rectangular channel receiver [70] (With Permission, License Number: 6 112 470 109 417).

exhibited excellent thermal uniformity, with temperature variations within 3°C along both the axial and lateral directions. Impressively, the system sustained a conversion efficiency of around 13.5% at an average cell temperature of just 15°C .

Kang et al. [71] proposed an innovative cooling strategy for HCPV systems by directly immersing solar cells in a PCM, creating a self-regulating cooling system that requires no external energy input. The experimental results demonstrated the system's effectiveness, with ethanol serving as the working fluid. At a high CR of 398.4 suns, the ethanol achieved a flow rate of up to $180.6\text{ kg/s}\cdot\text{m}^2$, while the surface heat transfer coefficient of the electric heating plate peaked at $23.49\text{ kW/m}^2\cdot\text{K}$. This enabled precise temperature control of the solar cells, maintaining the temperature within a narrow range of 87.3°C to 88.5°C . However, the phase transition of ethanol generated vapor bubbles, which negatively affected the electrical performance of the cells. These bubbles caused light scattering and absorption losses, leading to a 10.2% decrease in short-circuit current and a 7.3% drop in maximum power output.

4.4 | Microchannel Cooling

Microchannel cooling is an advanced thermal management technique that offers high-efficiency heat dissipation for CPV systems,

particularly under extreme solar flux conditions. This method involves embedding a network of microscale channels directly beneath or behind the solar cells, through which a cooling fluid, typically water or a dielectric liquid, is circulated. In addition, nanofluids have been greatly studied as a potential cooling fluid for such applications [72]. Due to the high surface-area-to-volume ratio of microchannels, this approach enables rapid heat transfer and maintains low temperature gradients across the cell surface, thereby minimizing thermal stresses and enhancing electrical efficiency. Microchannel cooling systems can also support higher CRs by preventing the formation of localized hot spots and maintaining uniform cell temperatures. However, challenges such as pressure drop, flow distribution, and potential clogging must be carefully addressed in the system design. Despite these considerations, microchannel cooling remains a promising solution for high-performance CPV systems where compactness, precise temperature control, and long-term reliability are critical.

Al Siyabi et al. [73] examined the variation of the heat sink's layer count, channel width, and height on its efficiency by building a 3D model, as shown in Figure 8. The system under study comprised a CPV MJSC module integrated with a multilayer microchannel heat sink. In this configuration, the assembly was positioned beneath a copper conductive layer, while a ceramic layer placed below the copper was used to provide electrical insulation. Performance was enhanced by adding more layers, which also decreased the temperature of the solar cells, reduced thermal resistance, and reduced the amount of electricity required for fluid pumping. For every arrangement, the maximum temperature recorded was lower than 69°C . The overall pressure drop was cut in half when switching from a single layer to a two-layer design; modest reductions were observed in three- and four-layer systems. When the channel height increased, the fluid pressure dropped, and the pumping power was significantly reduced. This led to a 3°C drop in solar cell temperature.

Radwan et al. [74] developed an innovative microchannel heat sink system incorporating two types of nanofluids: SiC-water and Al_2O_3 -water, to enhance cooling performance in CPV applications. As illustrated in Figure 9, the comprehensive model combined a detailed thermal analysis of the photovoltaic layers with thermo-fluid dynamics. The results showed that nanofluids significantly reduced solar cell temperatures compared to pure water, particularly under high CRs. The temperature reduction improved with increasing nanoparticle volume fraction, with

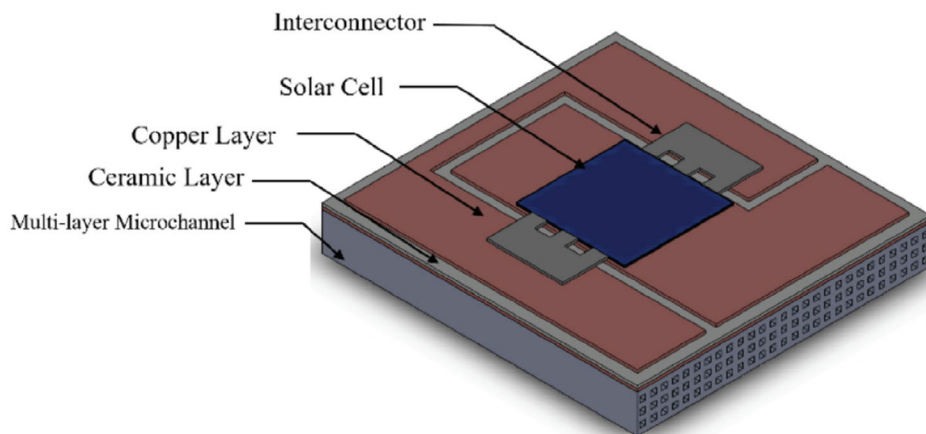


FIGURE 8 | A CPV receiver and multilayer microchannel heat sink [73] (With Permission, License Number: 6 114 021 207 137).

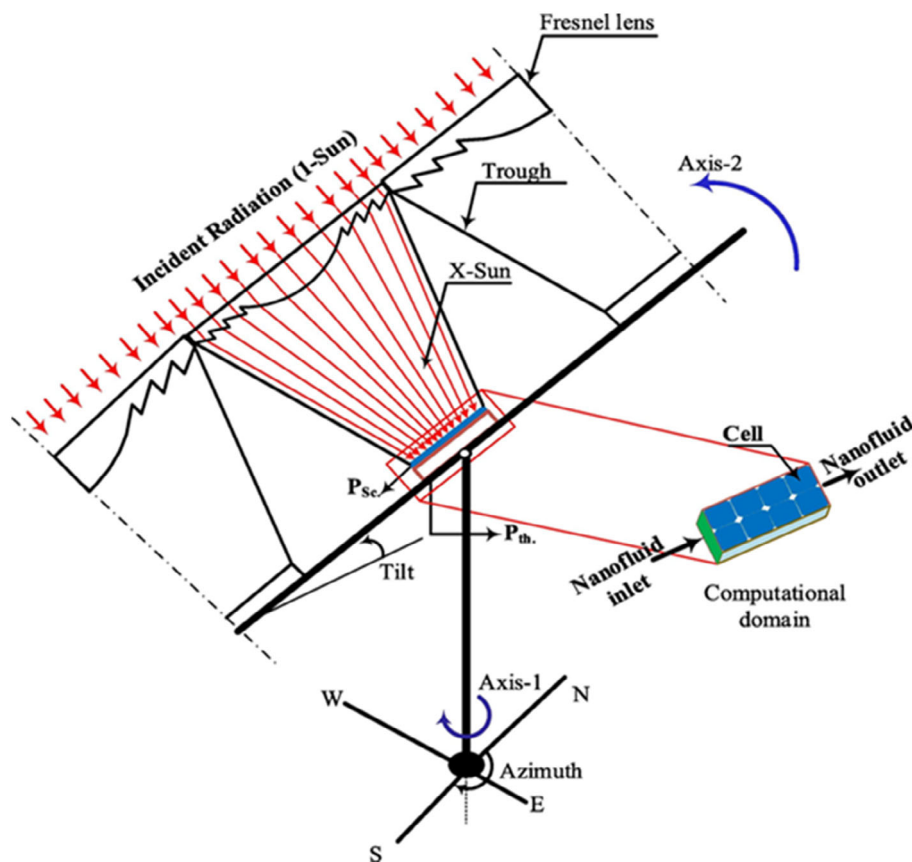


FIGURE 9 | The coordinate system and physical model of a CPV with nanofluid-based microchannel heat sink [74] (With Permission, License Number: 6 114 021 362 117).

SiC-water nanofluid exhibiting the highest cooling efficiency. At low Reynolds numbers and high concentration levels, nanofluids enhanced the electrical efficiency of the CPV system. However, at very high CRs, a decrease in thermal efficiency was observed. Both the Reynolds number and nanoparticle concentration contributed to a substantial rise in frictional losses, while net electrical power output varied with nanoparticle volume. Among the fluids tested, SiC-water outperformed both Al_2O_3 -water and pure water in thermal regulation. For example, at a Reynolds number of 10, a CR of 40, and a nanoparticle volume fraction of 4%, the cell temperatures were reduced to 53.4°C with pure water, 49.2°C with Al_2O_3 -water, and 48.0°C with SiC-water.

Thermal management in CPV systems using a double-layer microchannel heat sink (DL-MCHS) was investigated by Elqady et al. [75]. To simulate the thermal activity in the solar cell and fluid movement in the DL-MCHS, a 3D conjugate heat transfer model was created. The impact of headers on CFD calculations was the main topic of the study. Reliable thermo-hydraulic performance estimates were obtained from simulations with headers added, especially during counterflow operation. Without headers, calculations of pumping power were greatly affected. The temperature of the silicon wafer layer decreased when the volumetric flow rate increased because of the higher heat transfer coefficients inside the microchannel. The temperature differential between the header-in and header-out conditions was 12.36°C at a flow rate of 200 ml/h. The microchannel's internal pressure was reduced by adding headers, while pumping power was raised by 39.7%.

In order to address temperature nonuniformity problems in CPV receivers and reduce mismatch losses, Laguna et al. [76] developed a cooling device consisting of microfluidic cell matrices, each with a regulated coolant flow rate. In both uniform and nonuniform light profiles, the proposed device showed less temperature nonuniformities than conventional microchannel devices. The microfluidic cell matrix increased power generation by 9.7% at a steady 67°C. The CPV receiver significantly decreased temperature variations, with the largest variations being 34°C for conventional microchannels and 10°C for the microfluidic cell matrix. The power output of PV cells has been evaluated under various irradiance and temperature conditions, and a simplified model has been calibrated using the characteristic IV curve of the PV cells. Microchannel- and microfluidic-cell-cooled CPV receivers generate 72.6% and 79.7% as much power as electrically isolated cells under identical illumination and temperature conditions, respectively.

Rózsás et al. [77] demonstrated a more efficient method for incorporating microscale channel cooling networks into the backside metallization of CPV solar cells. The channels were fabricated by electroplating over a thick photoresist; improvements to the method have eliminated the residues and clogging caused by the photoresist, resulting in improved cooling performance. To avoid channel collapse or distortion following photoresist removal, the novel method uses an interim seed layer and patterned dashed lines. After considering the fluid dynamics and thermal effects, at a flow rate of 1.8 liters per hour, the thermal resistance was calculated to be 1.8 K/W. Maximum dissipation of 36 W, leading to a temperature increase to a maximum of 65°C, is predicted for

a 4 cm^2 concentrator solar cell operating at a concentration level of 100 suns.

4.5 | Jet Impingement Cooling

Jet impingement is an effective active thermal management technique for CPV systems, offering localized, high-intensity heat removal from solar cells exposed to extreme solar flux. This method involves directing a high-velocity fluid jet, typically air, water, or a dielectric liquid, onto the backside of the solar cell or its heat spreader surface, creating strong convective heat transfer at the point of impact. Due to its ability to rapidly dissipate heat from high-power-density regions, jet impingement cooling is particularly beneficial for maintaining uniform temperature distribution and preventing hot spots in high-concentration CPV modules. The cooling performance depends on factors such as jet velocity, nozzle-to-surface distance, fluid properties, and array configuration. Although it requires external pumping power and careful system integration, jet impingement provides precise temperature control and can significantly enhance the electrical efficiency and reliability of CPV systems operating under high irradiance conditions.

Awad et al. [78] developed a new jet impingement micro- and mini-channel heat sink design, shown in Figure 10, that has a heat spreader for CPV system thermal control. A thorough thermo-fluid model was created, validated experimentally, and numerically. At a CR of 20, the silicon layer temperature could be maintained at 87°C with a temperature uniformity of 3.7°C by employing the proposed design. The silicon layer temperature of the proposed design, in contrast, was 104°C with a temperature

uniformity of 15.5°C . The application of the heat spreader and jet impingement microchannel improved electrical efficiency and net power to 14.3% and 35.4 W, respectively, surpassing previous designs. By lowering silicon layer temperatures, raising the coolant's Reynolds number, and boosting net power and electrical efficiency, cooling performance was enhanced.

Abo Zahhad et al. [79] developed a specialized cooling strategy for HCPV systems aimed at optimizing the performance of triple-junction solar cells, as illustrated in Figure 11. The study employed 3D thermo-fluid models of four different confined jet impingement heat sink configurations, integrated with a detailed thermal model of the solar cell. It was concluded that increasing the coolant flow rate improved the electrical efficiency of the solar cells and achieved the necessary temperature uniformity across the cell surface. For instance, a single jet design reduced the temperature of an uncooled solar cell from a peak of 1360°C at a CR of 1000 suns to just 65°C , with thermal stress decreasing as the flow rate increased. The optimal exergy efficiency of 53.25% was achieved with a single jet configuration. Furthermore, the study provided practical implementation guidelines for HCPV/T (hybrid photovoltaic/thermal) systems that recover and utilize waste heat to enhance overall system performance.

Torbatinezhad et al. [80] examined the effects of mass flow rate and pin fin angle on the cooling efficiency of an HCPV system using a pin-finned jet impingement heat sink. The model aimed to reduce the surface temperature of triple-junction solar cells and enhance overall system performance. The results showed that increasing the pin fin angle from 0° to 40° significantly lowered the average cell temperature and improved thermal uniformity, leading to higher electrical efficiency. This improvement was primarily attributed to enhanced convective heat transfer resulting from the inclined fins. Additionally, increasing the mass flow rate of the coolant further reduced operating temperatures and promoted a more uniform temperature distribution, contributing to improved thermal and electrical performance. However, while higher mass flow rates and pin fin angles improved electrical exergy, they also resulted in reduced thermal and total exergy due to increased pressure drops and energy consumption. The study also found that increasing both parameters reduced the heat sink's thermal resistance but led to a decrease in the Nusselt number within the mini-centre channel. Despite these trade-offs, the system demonstrated optimal performance at a mass flow rate of 0.15 kg/s and a pin fin angle of 40° , achieving maximum electrical and thermal efficiencies of $\approx 29\%$ and 80% , respectively, with a peak exergy output of 2,760 W.

Tang et al. [81] utilized a manifold ultrathin micro pin-fin heat sinks (MUMPFHS) as a novel cooling technique for HCPV cells. A detailed 3D simulation model was created to calculate the energy, thermal, and exergetic performance of the system. The simulation results showed that the MUMPFHS design successfully lowered the temperature of the solar cell to 51°C from an intake temperature of 25°C . By reducing the average cell temperature and its nonuniformity, increasing the intake flow rate increased electrical efficiency. In order to attain a constant flow pattern, the study also added five different inlet/outlet plenum designs. This increased friction power but was countered by reduced solar cell temperatures, improving electrical power.

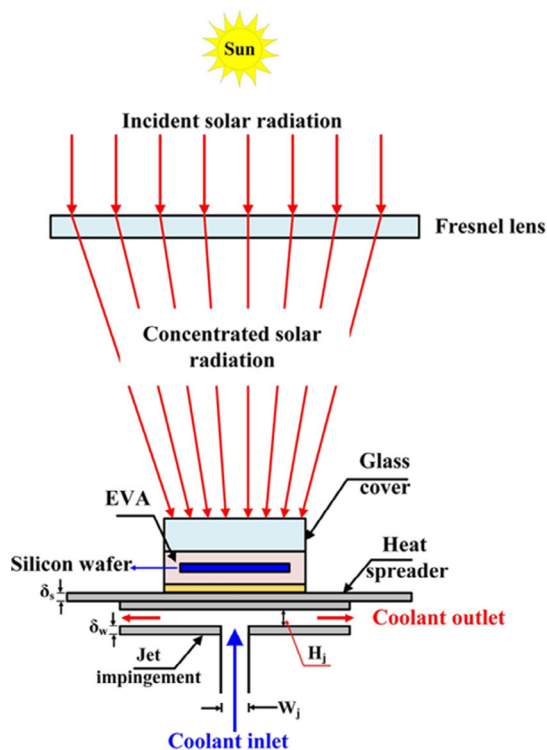


FIGURE 10 | The proposed heat management system that combines a CPV/jet impingement method with a heat sink system utilizing mini- and microchannels [78] (With Permission, License Number: 6 114 021 508 615).

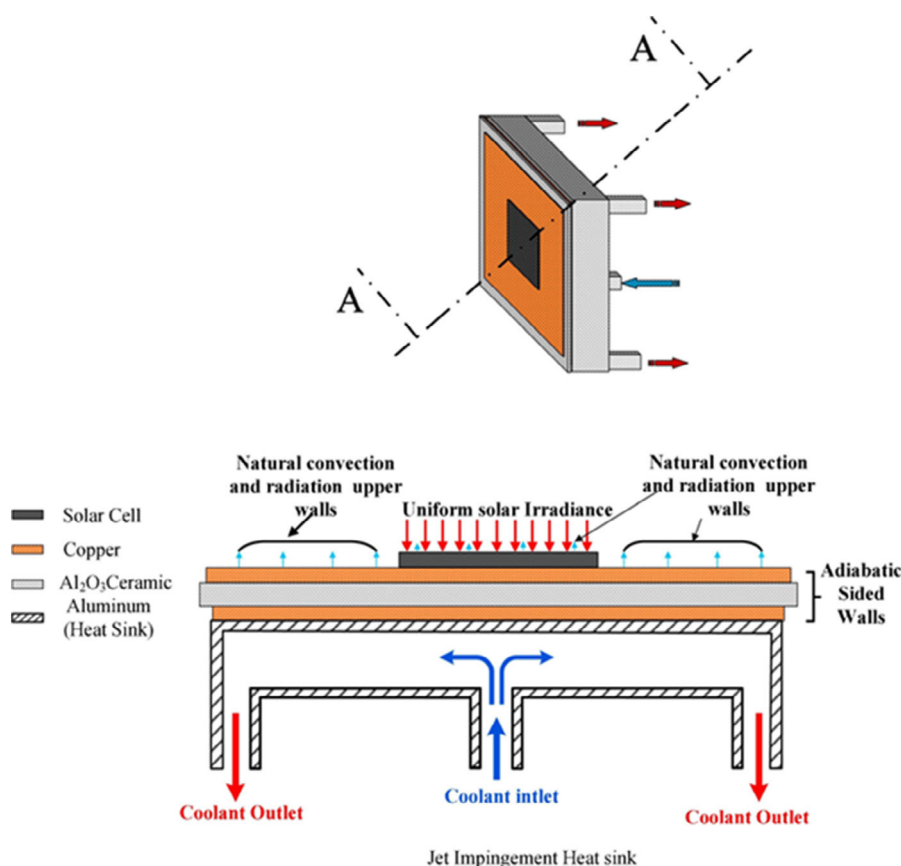


FIGURE 11 | A CPV with jet impingement heat sink assembly [79] (With Permission, License Number: 6 114 030 110 194).

4.6 | Comparative Assessment of CPV Cooling Techniques

While most CPV cooling studies primarily report temperature reduction and electrical efficiency enhancement, practical deployment requires a broader evaluation framework that also considers auxiliary energy consumption, system reliability, and adaptability to different operating scenarios. Table 5 summarizes the reviewed cooling techniques and their reported thermal and electrical outcomes; however, a comparative interpretation reveals important trade-offs that determine their suitability for real-world CPV applications.

Passive cooling approaches, including radiative cooling and PCM-based systems, are particularly attractive from an energy-consumption perspective, as they require little to no parasitic power. Radiative cooling strategies demonstrated acceptable temperature reductions and lifetime extensions without external energy input. This makes them highly suitable for remote, arid, or off-grid environments, where reliability and minimal maintenance are critical. Similarly, PCM-based cooling systems provide self-regulating thermal buffering under fluctuating irradiance conditions and offer improved thermal stability without pumping power. However, their scenario adaptability is limited by PCM melting temperature ranges and latent heat capacity, which must be carefully matched to local climatic conditions to avoid saturation during prolonged high-irradiance periods.

In contrast, active cooling techniques, such as microchannel and jet impingement cooling, enable operation at very high CRs and deliver superior temperature uniformity and control. These methods are highly adaptable to extreme solar flux conditions

and are well-suited for HCPV and CPV/T systems where waste heat recovery is desirable. However, this adaptability comes at the cost of increased energy consumption due to pumping power, pressure losses, and flow control requirements. In some cases, performance gains are partially offset by parasitic losses, especially at high Reynolds numbers.

From a reliability standpoint, passive systems generally offer higher long-term robustness due to their mechanical simplicity and absence of moving parts. Active systems, while thermally superior, introduce additional failure modes associated with pumps, seals, fouling, and fluid degradation, particularly in harsh outdoor environments. As evident from the presented literature, there is no single cooling technique that is universally optimal. For instance, passive cooling methods are best suited for low-to-medium CRs and energy-autonomous installations, whereas active cooling strategies are more appropriate for high-flux, space-constrained, or hybrid CPV/T systems where maximum power density is prioritized. Therefore, recent research efforts have focused on developing hybrid cooling systems that combine passive and active elements in advanced CPV designs to balance efficiency gains, energy consumption, operational reliability, and environmental adaptability.

5 | Applications and Integration of Advanced CPVs

Advanced applications and integration of CPV systems are expanding beyond traditional electricity generation, leveraging the high efficiency and optical flexibility to support hybrid and multifunctional energy solutions. One promising direction is the

TABLE 5 | Summary of reviewed studies on CPV cooling techniques.

Ref.	Year	Method	CR	Outcome
[58]	2020	Radiative	39	The GaSb cell's V_{oc} has increased by a relative 27%. It is also anticipated that the normal CPV cell's lifespan would increase by 4–15 times
[59]	2013	Radiative	820	Temperature drop of 10.5°C, with the addition of radiative cooling
[60]	2019	Radiative	13 to 35	Addition of radiative cooling decreased the temperature by 10°C, increased the V_{oc} by 5.7%, and the lifespan by 40%
[62]	2023	Radiative	200	The solar cell's temperature stays below 71.5°C at a CR of 200
[63]	2019	Radiative	39	The observed temperature decrease was 36°C, the rise in V_{oc} was around 0.08V, and extension of the cell's lifespan was 4–13 times
[65]	2018	PCM	3	The CPV-PCM/T system lowered the average temperature of the CPV modules from 24°C on average and by 28°C at peak times
[66]	2018	PCM	20	The temperature can be maintained below 72°C by employing a five-cavity design
[67]	2021	PCM	5 to 8	The new heat sink design with externally extended aluminum fins keeps the temperature at 82°C
[69]	2022	Immersed	—	Employing Water/ Al_2O_3 nanofluid as the cooling fluid decreased the top surface temperature by 10°C at the lowest Reynolds number of 15 000°C, and 8°C at the maximum Reynolds number of 30,000
[70]	2014	Immersed	9.1	A conversion efficiency of roughly 13.5% was kept at an average cell temperature of 15°C
[71]	2016	Immersed	398.4	The ethanol/bubble interaction resulted in a loss of light, which decreased the I_{sc} and P_{max} values by 10.2% and 7.3%
[73]	2017	Microchannel	500	The maximum temperature recorded was lower than 69°C
[74]	2016	Microchannel	40	The temperatures dropped by 4°C for Al_2O_3 -water, and 6°C for SiC-water
[75]	2020	Microchannel	20	The temperature differential between the header-in and header-out conditions was 12.36°C at a flow rate of 200 ml/hr
[76]	2017	Microchannel	350	The microfluidic cell matrix increased power generation by 9.7% at a steady 67°C
[77]	2020	Microchannel	100	Maximum dissipation of 36 W, leading to a temperature increase to a maximum of 65°C is predicted for a 4 cm ² concentrator solar cell operating at a concentration level of 100 suns
[78]	2020	Jet impingement	1 to 20	The application of the heat spreader and jet impingement microchannel improved electrical efficiency and net power to 14.3% and 35.4 W
[79]	2018	Jet impingement	1000	A single jet design lowered the temperature of the uncooled solar cell to 65°C at a coolant flow rate of 50 g/min
[80]	2021	Jet impingement	275 to 650	The greatest values of electrical and thermal efficiency were around 29% and 80%
[81]	2022	Jet impingement	1000	Lowered the temperature of the solar cell to 51°C from an intake temperature of 25°C at a flow rate of 3 kg/h

integration of CPVs with thermal energy recovery in hybrid CPV/T systems, which simultaneously produce electricity and capture waste heat for industrial processes, water heating, or desalination. CPV systems are also being explored in conjunction with hydrogen production technologies, such as photoelectrochemical cells, where their high photon flux can drive more efficient hydrogen evolution reactions. Additionally, advancements in lightweight concentrator optics, tracking mechanisms, and smart thermal management have enabled CPV deployment in space applications, remote off-grid areas, and building-integrated photovoltaics. Integration with digital monitoring, adaptive optics, and energy storage systems further enhances performance, reliability, and grid compatibility. These developments position advanced CPV systems as key enablers in diversified, high-performance renewable energy architectures.

5.1 | CPV and TEGs

A TEG is a heat recovery device that directly generates electricity from heat. It functions based on the Seebeck effect, which describes the phenomenon by which some materials produce an electric current in response to a change in temperature. TEGs have several uses, one of which is waste heat recovery, in which they are used to turn otherwise wasted heat from sources like industries or vehicles into energy. Although CPV and TEG technologies are not directly related, they can be coupled to improve power generation in some situations. TEGs could be employed to enhance the efficiency of CPV systems by capturing waste heat and converting it into energy [82]. Figure 12 illustrates a simplified schematic of the combined systems [83].

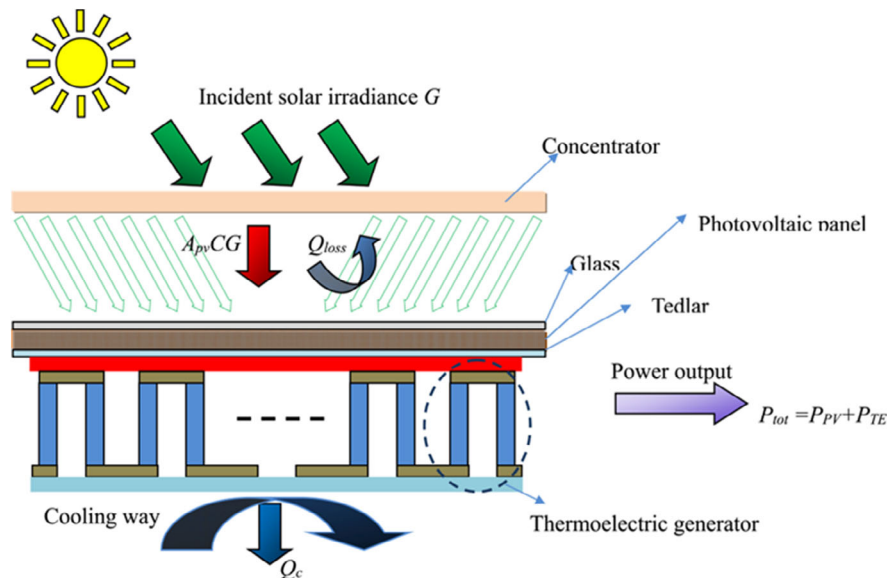


FIGURE 12 | A schematic diagram of a CPV-TEG system [83] (With Permission, License Number: 6 114 031 443 967).

Abdo et al. [84] optimized a photovoltaic-thermoelectric hybrid concentrator power-generating system as well as a microchannel heat sink (MCHS). A more recent design was contrasted with the MCHS positioned in between the two modules, with an older one that had the TEG attached to the rear of the CPV modules. The adjusted arrangement enhanced output power and decreased the average temperature. The new design produced 386 W/m^2 of total generated power, a conversion efficiency of 7.78%, and a temperature of 28°C at 5000 W/m^2 of solar radiation and 80 g/min of water flow. With an applied heat flux of $10,000 \text{ W/m}^2$, output power increased even further to 440 W/m^2 . With the solar cell and MCHS in direct contact, a higher CR could be achieved without going above the maximum operating temperature. This allowed the system to function efficiently at a maximum temperature of around 28°C and a water flow rate of about 80 g/min .

Cai et al. [83] examined the physical and operational factors, including solar radiation, CR, and the number of TEGs, affecting the CPV-TEG system's performance. The power output rose with solar radiation at first, but fell after a certain point. Depending on the CR, the maximum power output ranged from 58.85 W to 154.29 W . The Power output of the CPV system was increased by 14.22% by integrating TEGs. When solar radiation and CR increased, the energy destruction of CPV-TEG surpassed the power production of the system. Furthermore, power output and energy conversion efficiency are impacted by the number of TEGs, with larger CRs showing the greatest effects.

Rejeb et al. [85] examined the effects of four input parameters on system efficiency and established a statistical model to forecast the electrical efficiency of a CPV-TE. These variables include ambient temperature, leg height of TE, external load resistance, and the product of solar radiation and optical concentration. The researchers constructed a multiphysics numerical model that included radiative, conductive, and convective heat fluxes, Seebeck effect, and photoelectrical conversion using response surface methodology and finite element analysis in COMSOL. ANOVA was utilized to produce a quadratic regression model and evaluate the importance of every input factor, whereas the response surface method examined the outcomes of the

numerical model. Under optimal conditions, defined by particular values for solar radiation, temperature, load resistance, and TE module height, the CPV-TE system achieved a maximum electrical efficiency of 17.448%. The accuracy of the statistical model was validated with the numerical data. The electrical efficiency model demonstrated a high determination coefficient (0.9945).

Zhou et al. [86] integrated a metal-halide perovskite solar cell with a TEG to realize a CPV/TEG tandem device. TEG serves to alleviate thermal accumulation induced by concentrated solar irradiation. The resulting tandem system delivers a peak power conversion efficiency of 25.0% at 3 suns, exceeding the efficiency of the single perovskite solar cell by $\approx 4.7\%$. This study demonstrates that controlled heat extraction and reutilization in a perovskite solar cell/TEG tandem configuration can effectively harness excess thermal energy, offering a promising strategy to improve both efficiency and thermal stability of perovskite solar cells under low-concentration operation.

5.2 | Light Splitting CPV

This approach involves separating incoming sunlight into its constituent wavelengths and directing each spectral band to a solar cell that is specifically optimized to convert that portion of the spectrum. The efficiency can be significantly enhanced through the use of light-splitting techniques. Specialized optical filters and materials are used to achieve this spectral separation. Since different solar cells exhibit peak performance at different wavelengths, matching each band of light with the most suitable cell allows for more effective energy conversion. Through harnessing a broader range of the solar spectrum more efficiently, the overall performance and energy output of the CPV system are substantially improved.

An innovative CPV/T receiver design optimized for a linear flat mirror concentrator by Han et al. [87] maximized thermal and electrical outputs using solid and fluid-based absorptive filters. Modeling solar light propagation was made easier with the use of ray tracing software. Five fluid-based and two solid-based

filters were found using optical transmittance and merit functions. For the PV module, specific filter combinations produced surface illumination uniformity of above 96%. 33.67% of concentrated light, with 92.43% falling within the cell's spectrum response. A total of 49.88% of this energy was converted into electrical power by silicon CPV cells, with an electrical efficiency of 9.1% of the total incident power on the receiver.

A novel beam splitter and compact concentrator structure for a combined CPV/T system was presented by Wang et al. [88], which also provided information on the system architecture, design process, and optical filter design. The study tested simulated solar concentrating systems and examined beam splitting performance with respect to incident light angle, demonstrating exceptional concentration uniformity. To optimize the optical efficiency of the thermal utilization subsystem, the ideal configurations for solar receiver tube height and PV panel slope angle were established. Additionally, the study investigated the relation between optical performance and sun tracking precision and found that even a 1° angle deviation increased the optical efficiency to 66.2%. Thermodynamically, the CPV/T system outperformed traditional CPV systems in terms of PV conversion and overall energy efficiency, reaching its maximum output power at 356.0°C when operating.

A novel Spectral Splitting CPV/T, proposed by Han et al. [89], integrated a linear Fresnel reflector concentrator with an absorption optical filter. The optical performance of concentrator and hybrid receiver devices using ray tracing software was examined. Propylene glycol (PG) and HB650 filters produced a consistent light distribution and sent 72.15% of the spectral power in the 650–1075 nm range to the photovoltaic module. After the energy conversion performance was assessed, a simplified indoor test revealed that, in comparison to a PV-only system, a hybrid configuration with PG and HB650 filter boosted efficiency by 46.77% and economic value by 54%.

Using engineering equation solver theoretical simulations, Mohammadnia and Ziapour [90] studied an innovative hybrid system incorporating CPVs, a Stirling engine, and TEGs for solar energy conversion into electricity. By dividing the solar irradiance, a beam splitter directs longer wavelengths toward cavity absorbers to power Stirling engines and shorter wavelengths toward CPVs. Waste heat from the Stirling engine is transferred to cooling water using a novel TEG heat exchanger. A beam splitter cut-off wavelength of 875 nm yielded the highest energy efficiency of 24.57% at a CR of 431.4 sun, resulting in the generation of 38.34 kW of electrical power. Because of their principal contribution from solar radiation, CPVs have the best cost-performance. TEG modules made up only 0.32% of the system cost, but they provided 0.97% of the total electrical power. The TEG heat exchanger recovered ≈400 W of waste heat from the Stirling engine, increasing system efficiency without losing engine efficiency.

Zhu et al. [91] developed an innovative design for CPV/T systems that employs a solar collector tube as the SOE. The SOE served as a lens, heat collector, and spectrum beam splitter, similar to the POE of the system, which is a linear Fresnel lens. By lowering the average temperature of the PV module and increasing irradiation uniformity, this solution reduced production costs and problems while utilizing standard glass tubing. The system performed well within the range of 0–0.5°, achieving a reception angle of 0.59°. With a mean energy efficiency of 18.98%, which is made up of

13.76% electrical energy, 63.50% thermal energy, and 77.26% total energy, the system ensures consistent and effective output for a minimum of 6 h per day.

Kinoshit et al. [92] reported a panchromatic sensitizer, denoted DX3, which exhibits a broad photo-response extending into the near-infrared region up to ≈1100 nm and delivers a photocurrent density exceeding 30 mA cm⁻² under simulated AM 1.5 solar illumination. By integrating a DX3-based dye-sensitized solar cell with a perovskite solar cell that efficiently harvests visible light, a hybrid mesoscopic photovoltaic system employing spectral splitting was realized. This optically coupled architecture achieved a power conversion efficiency of 21.5%, demonstrating the effectiveness of spectral-splitting strategies for overcoming the long-wavelength absorption limitations of perovskite photovoltaics.

5.3 | CPV-Powered Desalination Systems

Using CPV technology to power desalination facilities has the potential to minimize the industry's financial and environmental impact. CPV technology may be employed in desalination systems in numerous ways. One of the most prevalent techniques for desalination, reverse osmosis (RO), might be powered by CPV. A RO system pumps water over a semipermeable membrane to remove salt and other contaminants from water. CPV technology might be implemented to run the pumps and other machinery required in the RO process. Powering thermal desalination systems is yet another use for CPV technology in the desalination industry. To create potable water from salty seawater, a thermal desalination system first generates steam, which is subsequently condensed. By using CPV technology to create the heat needed for making steam, dependency on fossil fuels might be diminished.

In order to desalinate brackish water, Wiesenfarth et al. [93] developed a CPV/T-based method that combined membrane distillation (MD) and RO technology, as shown in Figure 13. In addition to producing energy for desalination operations, the CPV/Thermal system has the capability to store and supply excess power to the grid. Although RO works well for desalination, the salt of the water increases the amount of energy it uses. As a secondary desalination technique, MD uses low-temperature heat from CPV/T; it is especially well-suited for brine concentration. When CPV/T and desalination are used together, brackish water with a salinity of 10–122.5 g/kg can be treated with a total recovery ratio of 92%. The CPV/T mirror dish has a maximum thermal efficiency of 79.2%, with minimal heat losses regardless of temperature gradient.

Khouya [94] examined the use of CPV/T in a hybrid desalination system that drives the processes of multiple effect distillation with thermal vapor compression (MED-TVC) and RO. Improved conversion rates were found for RO systems with increased feed pressure upon evaluation from different angles, especially for double and triple-stage RO. Higher feed pressure was shown in the study to result in lower permeate salinity and membrane module requirements; double-stage RO required the fewest modules. The steam ejector had moderate energy efficiency, and the preheater was the highest, with RO membranes exhibiting potential for improved performance. While MED-TVC systems showed specific thermal consumption of 24–160 kWh/m³, showing higher power requirements for MED-TVC, RO systems driven by CPV/T showed specific electrical consumption of 2.16–3.49 kWh/m³. The thermal

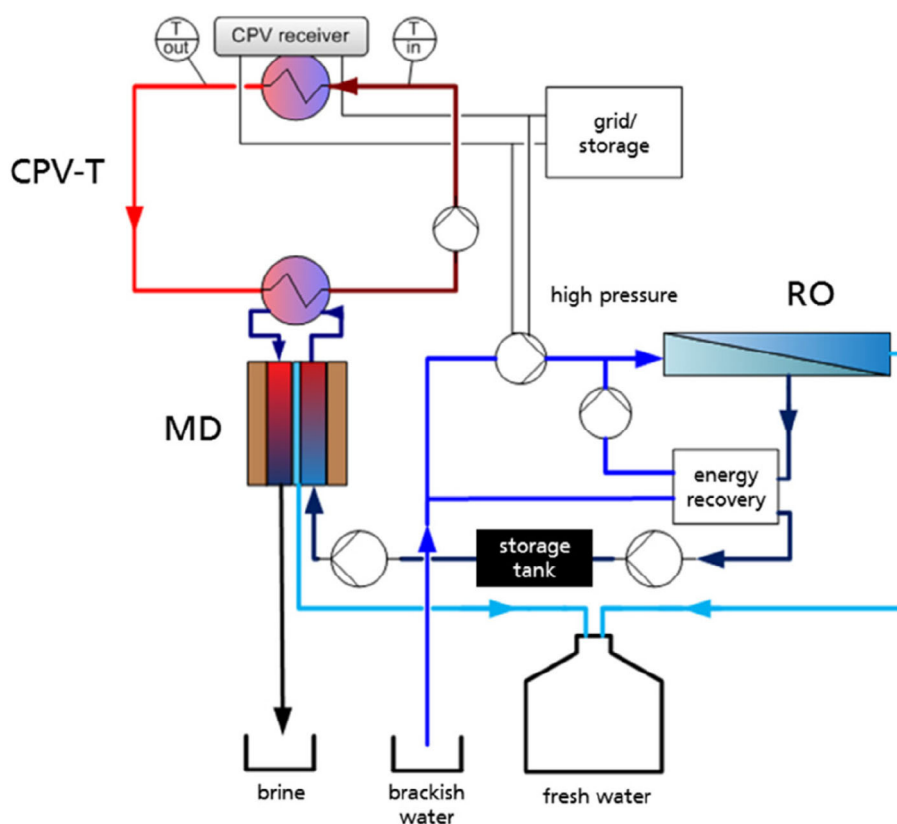


FIGURE 13 | CPV-powered desalination system [93] (With Permission, License Number: 6 114 030 321 874).

energy storage capacity of MED-TVC was enhanced by double the number of effects. The hybrid desalination plant showed a moderate carbon intensity of $0.67\text{--}0.87\text{ kg CO}_2/\text{m}^3$ and a low cost of water production (LCOW) ranging from $\$0.32$ to $\$0.7$ per m^3 . Based on these results, a double-stage RO integrated with a ten-effect MED-TVC was found to be the most efficient way to use heat and CPV/T electricity for desalination.

In order to generate electricity and freshwater in regions with limited land resources and freshwater shortages, Liang et al. [95] established an innovative system that integrates underwater solar CPV with MD, as shown in Figure 14. The device has a foldable solar concentrator that is suitable for use in maritime conditions and will not corrode from the salty sea air. The proposed system is composed of a transparent elastic membrane, a sealed cavity, a high-refractive index transparent cone, a solar cell, a cylindrical distillation membrane, and a condensation chamber. The inner surface of the sealed cavity is coated with a reflective material to enhance light reflection. Together, the elastic membrane, sealed cavity, and refractive cone form the underwater solar concentrator. The base of the cone is coupled to a solar cell, enabling direct conversion of concentrated sunlight into electricity. The cylindrical distillation membrane is a microporous, hydrophobic layer that separates liquid water from water vapor, allowing vapor to pass through while blocking liquid flow. Feed seawater is supplied on one side of the membrane, while the opposite side serves as the condensation chamber. Optimal performance values were found using optical simulations. An experimental study showed that the system operated consistently at any angle of incoming light. Overall, with a freshwater yield efficiency of 28.5% and a range of power outputs averaging 3.16% electrical

efficiency, the integrated system generated 67.8 g of freshwater. The overall energy consumption efficiency was 31.6%.

A CPV/T humidification and dehumidification desalination plant that uses intense sunlight to generate thermal energy and electricity was modelled mathematically by Anand and Srinivas [96]. Recovering waste heat through water flow facilitates the preservation of ideal PV cell temperatures. Using recovered heat energy, a humidification and dehumidification process produces desalination. The output of the system was calculated at various levels of solar radiation, and the flow rate of the coolant was adjusted correspondingly. With an energy utilization factor of 0.1593 and a gained output ratio of 3.01 at 800 W/m^2 and 300 kg/h , the maximum system efficiency for CPV/T is 88.40%. The generation of desalinated water peaks at 0.56 L/h at 57.5°C hot water temperature and 300 kg/h flow rate, while the production of power reaches 2.5 kW at 1000 W/m^2 and 400 kg/h flow rate.

Using an organic Rankine cycle (ORC) adsorption integration, Albaik et al. [97] investigated a combined CPV/T with the aim of producing electricity, cooling, heating, and water. Enhancing solar radiation absorption, the CPV system's optical efficiency was raised to 87.5% by applying a single layer of anti-reflective coating. Depending on the working conditions and the adsorber bed heat exchanger design, the integrated system provided multiple production types. The ORC's thermodynamic efficiency reached 11.6% when adsorption and ORC systems were coupled, giving the combined system a maximum efficiency of 68.47%. The output consisted of 79.17 liters of potable water per day, 13.08 kW of electrical power, and 2.06 kW of cooling power with solar power of 38.4 kW and thermal power of 20 kW from the CPV/T.

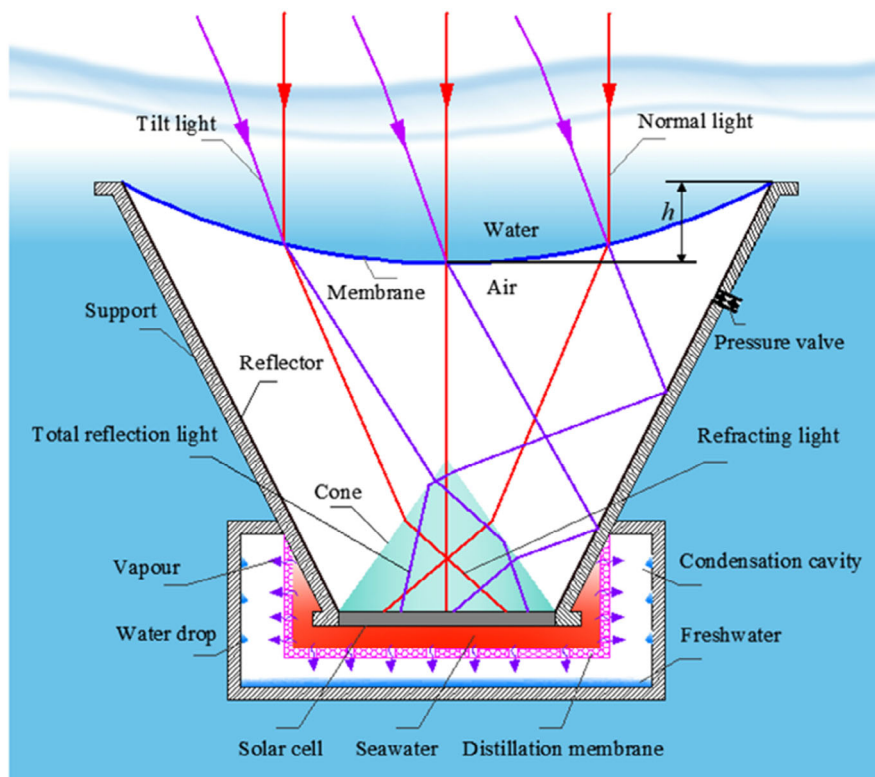


FIGURE 14 | Combined underwater solar CPV-MD [95] (With Permission, License Number: 6 114 030 539 261).

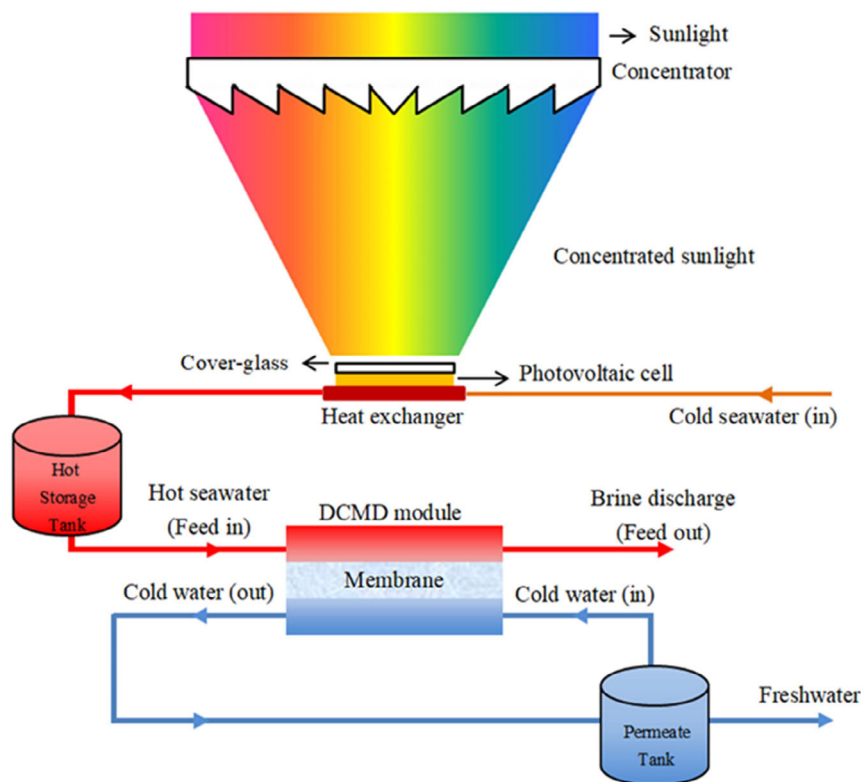


FIGURE 15 | A CPV-DCMD hybrid system [98] (With Permission, License Number: 6 114 030 783 816).

A method of recycling waste heat for the generation of potable water was proposed by Lai et al. [98] and combined CPV and direct contact membrane distillation (DCMD), as depicted in Figure 15. Studies on energy and exergy were carried out to

examine system performance, taking into account factors such as membrane properties, CPV temperature, and solar irradiation. When compared to solo CPV, the connected system demonstrated an increase of 35.37% in energy efficiency and 5.37% in

exergy efficiency. Higher feed flux or flow velocities were found to improve performance; however, higher permeate flux temperature had the opposite effect.

5.4 | Building Integrated CPV

Building integrated concentrated photovoltaic (BICPV) is a technology that merges the benefits of building integrated photovoltaic (BIPV) and CPV systems. The system is built into the structure, typically on the roof or in the facade, to produce energy and function as an aesthetic element. When compared to conventional PV systems, BICPV systems have many benefits. Since the system is incorporated into the building, it saves space and removes the need for additional mounting gear. Second, compared to conventional PV systems, the material requirements and environmental impact are lower due to the compact size and high efficiency of the solar cells. Also, the system outperforms conventional PV in terms of electricity production per area covered. Nevertheless, BICPV systems also have some drawbacks. First, they only function when exposed to sunlight, which might limit their use in some areas. Second, the system might be more expensive and complicated to maintain since it employs lenses or mirrors to focus sunlight and needs tracking devices to move with the sun throughout the day. Finally, if the structure is not made to deal with high temperatures, BICPV systems might cause issues. Besides these drawbacks, BICPV systems show substantial promise for decreasing a building's carbon footprint and producing sustainable electricity.

Window integration was suggested by Sharma et al. [99] using a unique window-integrated concentrated photovoltaic (WICPV) system. A comparative study was conducted utilizing the findings from the electrical and temperature characterizations that were carried out on all three cases: a PCM with a higher melting point (RT28HC), a PCM with a lower melting point (RT50), and circulating water cooling. Average maximum power is increased by 17%, 19%, and 25% with RT28HC, RT50, and flowing water, while electrical efficiency is improved by 15.9%, 17.6%, and 24.2%, respectively. The greatest measured average temperature drop inside the WICPV for each of the three cases is 6.0°C, 6.4°C, and 15.4°C, respectively. Figure 16 shows the proposed design in this study.

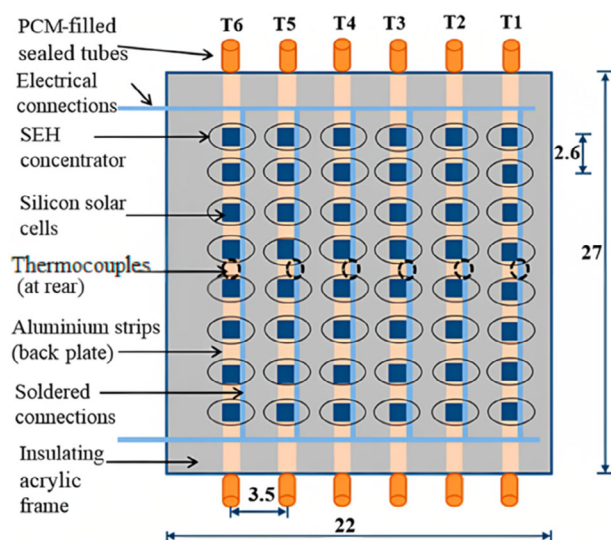


FIGURE 16 | A window-integrated CPV system [99] (Open Access).

Hussaina and Kim [100] investigated whether a point-focused CPV/T system could be more economically viable than standard gas and electricity heating systems for South Korean buildings during the harsh winter months. The apartment heating loads were evaluated using EnergyPlus modeling software, and it was discovered that the CPV/T system could provide all heating requirements for several months without requiring additional grid electricity. When heating apartments with the CPV/T system vs electricity and natural gas, comparative life cycle savings of \$2,891 and \$2,370 were computed, with corresponding payback periods of 12 and 15 years during the system's expected 25-year service life. The efficiency of the CPV/T system was shown to be impacted by operational changes, with thermal fluid flow rate having a major effect on electrical efficiency. The CPV/T system proved to have high thermal and electrical efficiency of 65.2% and 18.5%, respectively, even in cold weather, making it a useful heating choice.

Zhu et al. [101] investigated the heat transmission performance of a double-skin facade (DSF) with high-concentration solar cells, or CPV-DSF. A thermal resistance model, an energy balancing formula, and a method for figuring out the facade's overall heat transfer coefficient were created. The channel and vent width were optimized, and the distribution of solar radiation using TracePro and Fluent software was studied. It was discovered that the best values for channel width and ventilation outlet width to maximize exhaust heat, internal airflow, and solar radiation transmittance were 0.5 and 0.2 m, respectively. With an effective thermal energy exchange between the interior and exterior, the CPV-DSF demonstrated a total heat transfer coefficient of 1.179 W/m²·K under these conditions. The facade's 0.395 m³/s vent flow rate and average wind speed of 0.52 m/s guarantee effective ventilation for preserving indoor air quality. By reducing thermal degradation, exhaust heat removal at 143.70 W improves cell performance and longevity. Average CPV cell temperatures are around 15°C lower than in the absence of ventilation.

A CPV/T module for building louvers was presented by Liang et al. [102], using vertical space to capture solar radiation for energy conversion. This module, which has a high-efficiency photovoltaic-thermal working range of 20°–70° and is functional even at low solar angles of 20°, blends a PV/T module and a focusing blade for best efficiency. In order to reduce the need for cooling, the blade collects waste heat by using thermal fluid to guide sunlight into solar cells. The geometrical CR reached a maximum of 2.96 at a 20° incidence angle, according to optical simulations, and remained above 2 for incidence angles ranging from 12.5° to 52.5°. Following testing for a full day, the CPV/T module demonstrated efficiency above 55% for 5 h without the need for monitoring. It peaked at 56.8% total efficiency at a 20° tilt incidence angle, with matched electrical and thermal efficiencies of 4.4% and 52.4%, respectively. During an extended testing duration, the CPV/T module produced more than 2.20 times the power of a single solar cell, underscoring its significant potential for energy generation.

5.5 | High Concentrated CPV

HCPV is characterized by its ability to concentrate sunlight over a small area (at high concentration levels), hence increasing the power output of individual solar cells. To do this, a tracking

device that follows the sun's motion during the day is combined with lenses or mirrors. Because of the high solar insolation, less photovoltaic material may be used to produce the same quantity of power, which can assist in bringing down the overall cost of the technology. However, unlike conventional solar panels, high-concentration CPV systems require accurate alignment and tracking and may be more vulnerable to weather fluctuations.

Talavera et al. [103] evaluated the HCPV technology's economic viability in Spain by calculating the LCOE. The results were illustrated with a series of maps showing that, when cost-of-ownership is taken into account, HCPV is starting to become competitive in some areas. HCPV systems obtained LCOEs ranging from 0.081 to 0.18 €/kWh in places with DNI in 2013 between 2221 and 982 kWh/(m²·year). With ongoing market growth, projections indicate that by 2020, LCOEs for HCPV systems might fall to 0.035–0.080 €/kWh. The study also found that, throughout all of Spain, HCPV is not intrinsically less lucrative than traditional PV systems, with the level of yearly direct and global irradiation in a particular area having a significant impact on the technology selection. Spain's southwest and northeast are the best places to install HCPV systems, whilst the country's northern region is better suited for traditional PV installations.

Using two p–n junctions connected in series by a highly doped tunnel diode with lateral metallic contacts, Fernández et al. [104] presented an innovative solar cell design designed for ultra-high light concentration conditions exceeding 2000 suns. To optimize solar spectrum absorption, especially with III–V semiconductors, this novel architecture makes use of tunnel diode connections and direct band-gap semiconductor compounds. The efficiency of this design was assessed using TCAD simulations at concentration levels up to 15 000 suns, showing that the prototype's efficiency of 28.4% was constant without declining with concentration. According to simulations, efficiency increased linearly with logarithmic concentration increases, from 22.5% at 1 sun to 28.4% at 15 000 suns. This suggests that ultra-high concentration systems may be able to achieve cost savings and efficiency gains.

Micheli et al. [105] developed the first application of the least-material technique in CPV to maximize heat dissipation and minimize weight, introducing a novel approach to designing heat sinks optimized for CPV systems operating at concentrations exceeding 2,000 suns. The goal is to lower total technological costs while keeping multijunction cell temperatures within their operating range. The simulation results show that even at very high concentrations, passive cooling is still effective. An aluminum heat sink with 8 fins, measuring 11 cm wide and 6 cm high, is able to maintain operational temperature ranges for a 3 mm x 3 mm MJSC operating at a CR of 4000 suns. Using this method, the authors predicted that the construction costs for ideal heat sinks would vary from 0.10 to 0.18\$/Wp for concentrations between 1000x to 8000x.

5.6 | Digital Integration in CPV

Digitalization is reshaping energy technologies, including data-driven tools such as AI, Internet of Things sensing, and digital twins, which are being used to improve prediction, control, reliability, and lifetime operation. In solar energy, more specifically, AI and IoT have been widely applied for irradiance forecasting, fault detection, and smart solar-tracking, and recent work has

demonstrated effective AI-driven tracking and sensor-based control strategies for PV systems. On the other hand, the literature that investigates digital integration specifically for CPV and CPV/T is still smaller than the broader PV/solar body of work. Recent research studies show promising demonstrations, where several intelligent methods developed for nonconcentrated PVs, such as advanced solar-tracking algorithms, digital-twin-based thermal-management optimization, and IoT-enabled monitoring, are directly transferable to CPV systems and can accelerate CPV research and deployment.

One of the earliest and most extensive applications of digital tools in CPV comes from artificial neural networks (ANNs) used for electrical and environmental modeling. Because CPV devices exhibit nonlinear interactions between optics, solar cells, and environmental variables, traditional analytical models often fall short. Almonacid et al. [106] highlighted how ANNs have been successfully applied to characterize the electrical performance of LCPV and HCPV devices, estimate MJSC temperature from atmospheric inputs, and model spectral effects. Their findings confirm that ANN-based models provide high-accuracy predictions and can simulate MJSC behavior without intensive physical measurements. Environmental variability strongly influences CPV/T output, making accurate prediction of solar radiation, ambient temperature, and spectral conditions essential. Masood et al. [107] demonstrated the feasibility of applying AI tools such as ANNs, genetic algorithms, and composite models to predict solar insolation and ambient temperature with high precision. These predictions can be directly incorporated into thermal-management strategies and real-time control algorithms, forming the basis for intelligent CPV/T operation. Recent efforts extend beyond simple prediction to real-time optimization of CPV/T operation using deep learning, reinforcement learning, and hybrid intelligent algorithms. Kolamroudi et al. [108] showed how these techniques enable dynamic cooling strategies, precision solar tracking, and fault detection. Reported improvements include more than 35% enhancement in thermal stability, up to 85% reduction in mirror misalignment, and R^2 values exceeding 0.99 for CPV/T energy-output prediction.

Another emerging direction is the hybridization of statistical and ML models to predict CPV system output under future climate scenarios. The study conducted by Nwokolo et al. [109] demonstrated that integrating SARIMA, CARIMA, and logistic distribution models yields a high-accuracy hybrid predictor for both single-axis and dual-axis tracked CPV systems. With R^2 values around 0.98, this approach outperformed conventional multilayer perception and radial basis function neural networks. The model was successfully validated across five African climatic zones, showing the potential for climate-adaptive CPV performance forecasting. ML has also been incorporated directly into real-time improvements of CPV/T systems. The ML-assisted mirror-tracking strategy proposed in [110] combines proportional control with online regression-based ML correction to dynamically refine mirror orientation. Experimental validation on a CPV/T testbed demonstrated nearly 500% improvement in thermal-power stability compared to conventional mirrorless configurations.

Beyond AI algorithms, IoT-based tracking systems are becoming increasingly important for CPV implementations. An IoT-embedded concentrator prototype demonstrated a 15% efficiency improvement over conventional PV systems by integrating IoT sensors for solar tracking, monitoring, and real-time actuation

[111]. Meanwhile, the knowledge-based tracker presented by Canada-Bago et al. [112] used fuzzy rule-based systems implemented via IoT hardware to overcome practical inaccuracies in real-world HCPV installations. Their system mainly compensated for manufacturing misalignment, installation tolerances, sensor uncertainty, and mechanical imperfections.

6 | Conclusion

CPV systems maximize power generation per unit area by concentrating sunlight into a small area using specialized optics and tracking devices. For this reason, CPV systems can outperform conventional solar panels in terms of efficiency. In comparison to the efficiency of conventional solar panels, which is typically between 15 and 20%, CPV systems may reach up to 40%. This paper presented a comprehensive review of CPV systems, with a focus on various cooling technologies and applications such as desalination, waste heat recovery, light splitting, BICPV, and HCPV. The main findings can be summarized as follows:

- Radiative cooling has shown significant improvements in the lifespan and V_{oc} of CPV cells, with temperature reductions up to 36°C. PCM cooling has also been effective, with temperature drops up to 28°C. Immersion cooling using nanofluids and ethanol/bubble interaction has shown mixed results, with some loss of light in the latter. Microchannel cooling has been effective in maintaining temperatures below 69°C. Jet impingement cooling has achieved efficiencies up to 29% and temperature reductions to 51°C.
- Multiple methods like MED, RO, humidification, and dehumidification have been used. These methods have shown promising results, with recovery ratios up to 92% and efficiencies up to 68.47%.
- TEGs have been used for WHR, increasing the power output of the CPV system by up to 14.22% and achieving a maximum electrical efficiency of 17.448%.
- Different concentrators and beam splitters have been used to increase the overall optical efficiency up to 66.2% and energy efficiency up to 24.57%.
- BICPV combined with different cooling methods resulted in increased power and improved electrical efficiency. The CPV/T system has shown high thermal and electrical efficiency of 65.2% and 18.5%.
- HCPV systems have obtained LCOE ranging from 0.081 to 0.18 €/kWh. Efficiency increased with the increase in concentration, from 22.5% at 1 sun to 28.4% at 15 000 suns.

Author Contributions

Abdul Ghani Olabi: conceptualization (lead), supervision (lead), writing – original draft (supporting), and writing – review and editing (equal). **Adnan Alashkar:** formal analysis (equal), writing – original draft (equal), and writing – review and editing (equal). **Montaser Mahmoud:** investigation (lead), writing – original draft (equal), and writing – review and editing (equal). **Ayman Mdallal:** formal analysis (equal), writing – original draft (equal), and writing – review and editing (equal). **Malek Kamal Rabaia:** data curation (lead), writing – original draft (supporting), and writing – review and editing (supporting). **Abdul Hai Alami:** visualization

(lead), writing – original draft (supporting), and writing – review and editing (supporting). **Mahmoud Elgendi:** project administration (lead), writing – original draft (supporting), and writing – review and editing (equal). **Mohammad Ali Abdelkareem:** methodology (lead), writing – original draft (equal), and writing – review and editing (supporting).

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT and Grammarly to rephrase sentences and enhance language. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Acknowledgements

Open access publishing facilitated by United Arab Emirates University, as part of the Wiley - United Arab Emirates University agreement.

Funding

The author has nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

References

1. M. F. Hossain, “The Availability of Global Solar Energy,” in *Sustainable Global Development in Environment, Energy, Building, and Infrastructure*, ed. M. F. Hossain (Springer Nature Switzerland, 2025), 55–67, https://doi.org/10.1007/978-3-031-84429-4_6.
2. P. M. Tembo and V. Subramanian, “Current Trends in Silicon-Based Photovoltaic Recycling: A Technology, Assessment, and Policy Review,” *Solar Energy* 259 (2023): 137–150, <https://doi.org/10.1016/j.solener.2023.05.009>.
3. I. Ganesh, “Technologies Available Today to Generate Electricity from Sunlight and Their Current Status,” in *Harvesting Solar Energy*. Green Energy and Technology (Springer Singapore, 2025), 17–55, https://doi.org/10.1007/978-981-96-2321-1_2.
4. D. del Col, M. Bortolato, A. Padovan, and M. Quaggia, “Experimental and Numerical Study of a Parabolic Trough Linear CPVT System,” *Energy Procedia* 57 (2014): 255–264, <https://doi.org/10.1016/j.egypro.2014.10.030>.
5. F. L. Rashid, M. A. Al-Obaidi, A. J. Mahdi, and A. Ameen, “Advancements in Fresnel Lens Technology across Diverse Solar Energy Applications: A Comprehensive Review,” *Energies (Basel)* 17, no. 3 (2024): 569, <https://doi.org/10.3390/en17030569>.
6. A. Kharaghani, M. Zamen, and A. A. Nejad, “Optimal Design, Construction and Evaluation of Symmetric Compound Parabolic Concentrator Integrated with Photovoltaics System (CPC-PV),” *Solar Energy* 277 (2024): 112756, <https://doi.org/10.1016/j.solener.2024.112756>.
7. X. Li, K. Li, Y. Sun, et al., “Comprehensive Investigation of a Building Integrated Crossed Compound Parabolic Concentrator Photovoltaic Window System: Thermal, Optical and Electrical Performance,” *Renew Energy* 223 (2024), 119791, <https://doi.org/10.1016/j.renene.2023.119791>.
8. X. Zhang, S. Jiang, Z. Lin, Q. Gui, and F. Chen, “Model Construction and Performance Analysis for Asymmetric Compound Parabolic Concentrator with Circular Absorber,” *Energy* 267 (2023), 126597, <https://doi.org/10.1016/j.energy.2022.126597>.

9. F. Masood, P. Nallagownden, I. Elamvazuthi, J. Akhter, and M. A. Alam, A New Approach for Design Optimization and Parametric Analysis of Symmetric Compound Parabolic Concentrator for Photovoltaic Applications," *Sustainability* 13, no. 9 (2021): 4066, <https://doi.org/10.3390/su13094066>.
10. Q. Xuan, G. Li, H. Yang, et al., "Performance Evaluation for the Dielectric Asymmetric Compound Parabolic Concentrator with Almost Unity Angular Acceptance Efficiency," *Energy* 233 (2021): 121065, <https://doi.org/10.1016/j.energy.2021.121065>.
11. W. P. Wong, R. Walvekar, M. Vaka, M. Khalid, and N. M. Mubarak, A Review on CSP and CPVT Technologies: Energy, Economic Analysis and Recycling towards Circular Economy," *ECS Journal of Solid State Science and Technology*, 12, no. 3 (2023): 01102, <https://doi.org/10.1149/2162-8777/acc4e6>.
12. M. Khamooshi, H. Salati, F. Egelioglu, A. H. Faghiri, J. Tarabishi, and S. Babadi, "A Review of Solar Photovoltaic Concentrators," *International Journal of Photoenergy* 2014 (2014): 958521, <https://doi.org/10.1155/2014/958521>.
13. M. A. Fazal and S. Rubaiee, "Progress of PV Cell Technology: Feasibility of Building Materials, Cost, Performance, and Stability," *Solar Energy* 258 (2023): 203–219, <https://doi.org/10.1016/J.SOLENER.2023.04.066>.
14. NREL, "Best Research-Cell Efficiency Chart," accessed September 18, 2025, <https://www.nrel.gov/pv/cell-efficiency>.
15. I. S. E. Fraunhofer, "Fraunhofer ISE Develops the World's Most Efficient Solar Cell with 47.6% Efficiency," 2025, accessed September 18, <https://www.ise.fraunhofer.de/en/press-media/press-releases/2022/fraunhofer-ise-develops-the-worlds-most-efficient-solar-cell-with-47-comma-6-percent-efficiency.html>.
16. R. McConnell and V. Fthenakis, "Concentrated Photovoltaics," in *Third Generation Photovoltaics* (InTech, 2012), <https://doi.org/10.5772/39245>.
17. L. C. Spencer, "A Comprehensive Review of Small Solar-Powered Heat Engines: Part I. A History of Solar-Powered Devices up to 1950," *Solar Energy* 43, no. 4 (1989): 191–196, [https://doi.org/10.1016/0038-092X\(89\)90019-4](https://doi.org/10.1016/0038-092X(89)90019-4).
18. G. Sala and A. Luque, "Past Experiences and New Challenges of PV Concentrators," in *Concentrator Photovoltaics* (Springer Berlin Heidelberg, 2007), 1–23, https://doi.org/10.1007/978-3-540-68798-6_1.
19. R. Sinton, Y. Kwark, and R. M. Swanson, "VIB-1 Twenty-Three-Percent Efficient Si Point Contact Concentrator Solar Cell," *Ieee Transactions on Electron Devices* 32, no. 11 (1985): 2553–2554, <https://doi.org/10.1109/T-ED.1985.22376>.
20. R. Winston and H. Hinterberger, "Principles of Cylindrical Concentrators for Solar Energy," *Solar Energy* 17, no. 4 (1975): 255–258, [https://doi.org/10.1016/0038-092X\(75\)90007-9](https://doi.org/10.1016/0038-092X(75)90007-9).
21. P. Benitez, M. Hernandez, R. M. Arroyo, J. C. Minano, and F. Munoz, "New Nonimaging Static Concentrators for Bifacial Photovoltaic Solar Cells," 3781 (1999): 22–29, <https://doi.org/10.1117/12.368256>.
22. I. Tobías, A. Luque, and J. C. Miñano, "Light Confining Cavities for GaAs and Si Solar Cells," *International Journal of Solar Energy* 10, no. 3-4 (1991): 185–204, <https://doi.org/10.1080/01425919108941462>.
23. S. El Himer, S. El Ayane, S. El Yahyaoui, J. P. Salvestrini, and A. Ahaitouf, "Photovoltaic Concentration: Research and Development," *Energies* 13, no. 21 (2020): 5721, <https://doi.org/10.3390/en13215721>.
24. A. T. Raisa, S. N. Sakib, M. J. Hossain, K. A. Rocky, and A. Kowsar, "Advances in Multijunction Solar Cells: An Overview," *Solar Energy Advances* 5 (2025): 100105, <https://doi.org/10.1016/j.seja.2025.100105>.
25. M. A. Green, K. Emery, Y. Hishikawa, W. Warta, and E. D. Dunlop, "Solar Cell Efficiency Tables (version 42)," *Progress in Photovoltaics: Research and Applications* 21, no. 5 (2013): 827–837, <https://doi.org/10.1002/PIP.2404>.
26. A. Baiju and M. Yarema, "Status and Challenges of Multi-Junction Solar Cell Technology," *Frontiers in Energy Research* 10 (2022): 971918, <https://doi.org/10.3389/fenrg.2022.971918>.
27. A. O. Ali, A. T. Elgohr, M. H. El-Mahdy, et al., "Advancements in Photovoltaic Technology: A Comprehensive Review of Recent Advances and Future Prospects," *Energy Conversion and Management: X* 26 (2025), 100952, <https://doi.org/10.1016/j.ecmx.2025.100952>.
28. J. H. Ermer, R. K. Jones, P. Hebert, et al., "Status of C3MJ+ and C4MJ Production Concentrator Solar Cells at Spectrolab," *IEEE Journal of Photovoltaics* 2, no. 2 (2012): 209–213, <https://doi.org/10.1109/JPHOTOV.2011.2180893>.
29. M. A. Green, E. D. Dunlop, D. H. Levi, J. Hohl-Ebinger, M. Yoshita, and A. W. Y. Ho-Baillie, "Solar Cell Efficiency Tables (version 54)," *Progress in Photovoltaics: Research and Applications* 27, no. 7 (2019): 565–575, <https://doi.org/10.1002/PIP.3171>.
30. NREL, "NREL demonstrates 45.7% Efficiency for Concentrator Solar Cell," accessed August 25, 2025, <https://phys.org/news/2014-12-nrel-efficiency-solar-cell.html>.
31. K. Sasaki, T. Agui, K. Nakaido, N. Takahashi, R. Onitsuka, and T. Takamoto, "Development of InGaP/GaAs/InGaAs Inverted Triple Junction Concentrator Solar Cells," *AIP Conference Proceedings* 1556, no. 1 (2013): 22, <https://doi.org/10.1063/1.4822190>.
32. V. Sabnis, H. Yuen, and M. Wiemer, "High-Efficiency Multijunction Solar Cells Employing Dilute Nitrides," *AIP Conference Proceedings* 1477, no. 1 (2012): 14, <https://doi.org/10.1063/1.4753823>.
33. J. F. Geisz, A. Duda, R. M. France, et al., "Optimization of 3-Junction Inverted Metamorphic Solar Cells for High-Temperature and High-Concentration Operation," *AIP Conference Proceedings* 1477, no. 1 (2012): 44, <https://doi.org/10.1063/1.4753830>.
34. P. Chiu, S. Wojtczuk, X. Zhang, C. Harris, D. Pulver, and M. Timmons, "42.3% Efficient InGaP/GaAs/InGaAs Concentrators Using Bifacial Epigrowth," *Conference Record of the 37th IEEE Photovoltaic Specialists Conference* (2011): 000771–000774, <https://doi.org/10.1109/PVSC.2011.6186067>.
35. R. King, A. Boca, W. Hong, et al., "Band-Gap-Engineered Architectures for High-Efficiency Multijunction Concentrator Solar Cells," poster presented at the 24th European Photovoltaic Solar Energy Conference and Exhibition, September 2009, <https://doi.org/10.4229/24thEUPVSEC2009-1AO.5.2>.
36. W. Guter, J. Schöne, S. P. Philipps, et al., Current-Matched Triple-Junction Solar Cell reaching 41.1% Conversion Efficiency under Concentrated Sunlight," *Applied Physics Letters*, 94, no. 22 (2009): 223504, <https://doi.org/10.1063/1.3148341>.
37. F. E. Cherif and H. Sammouda, "Strategies for High Performance Perovskite/c-Si Tandem Solar Cells: Effects of Bandgap Engineering, Solar Concentration and Device Temperature," *Optical Materials* 106 (2020): 109935, <https://doi.org/10.1016/j.optmat.2020.109935>.
38. S. P. Philipps, A. W. Bett, K. Horowitz, and S. Kurtz, "Current Status of Concentrator Photovoltaic (CPV) Technology," Fraunhofer Institute for Solar Energy Systems (ISE) and National Renewable Energy Laboratory (NREL), 2015, accessed September 18, 2025. [Online]. <https://docs.nrel.gov/docs/fy16osti/65130.pdf>.
39. J. C. Miñano, P. Benítez, P. Zamora, M. Buljan, R. Mohedano, and A. Santamaría, "Free-Form Optics for Fresnel-Lens-Based Photovoltaic Concentrators," *Optics Express* 21, no. S3 (2013): A494, <https://doi.org/10.1364/OE.21.00A494>.
40. H. J. Queisser, "Slip Patterns on Boron-Doped Silicon Surfaces," *Journal of Applied Physics* 32, no. 9 (2004): 1776, <https://doi.org/10.1063/1.1728435>.
41. R. Winston, J. C. Miñano, and P. Benítez, *Nonimaging Optics* (Elsevier, 2005), <https://doi.org/10.1016/B978-0-12-759751-5.X5000-3>.
42. R. Baños, F. Manzano-Agugliaro, F. G. Montoya, C. Gil, A. Alcayde, and J. Gómez, "Optimization Methods Applied to Renewable and

- Sustainable Energy: A Review,” *Renewable and Sustainable Energy Reviews* 15, no. 4 (2011): 1753–1766, <https://doi.org/10.1016/J.RSER.2010.12.008>.
43. Y. C. Chen and H. W. Chiang, “Design of the Secondary Optical Elements for Concentrated Photovoltaic Units with Fresnel Lenses,” *Applied Sciences* 5, no. 4 (2015): 770–786, <https://doi.org/10.3390/APP5040770>.
 44. M. Victoria, C. Domínguez, I. Antón, and G. Sala, “Comparative Analysis of Different Secondary Optical Elements for Aspheric Primary Lenses,” *Optics Express* 17, no. 8 (2009): 6487–6492, <https://doi.org/10.1364/OE.17.006487>.
 45. L. Fu, R. Leutz, and H. P. Annen, “Secondary Optics for Fresnel Lens Solar Concentrators,” in *Nonimaging Optics: Efficient Design for Illumination and Solar Concentration VII*, Vol. 7785 (SPIE, 2010), 81–86, <https://doi.org/10.1117/12.860438>.
 46. T. Hornung, M. Steiner, and P. Nitz, “Estimation of the Influence of Fresnel Lens Temperature on Energy Generation of a Concentrator Photovoltaic System,” *Solar Energy Materials and Solar Cells* 99 (2012): 333–338, <https://doi.org/10.1016/J.SOLMAT.2011.12.024>.
 47. D. C. Miller and S. R. Kurtz, “Durability of Fresnel Lenses: A Review Specific to the Concentrating Photovoltaic Application,” *Solar Energy Materials and Solar Cells* 95, no. 8 (2011): 2037–2068, <https://doi.org/10.1016/J.SOLMAT.2011.01.031>.
 48. M. Buljan, J. Mendes-Lopes, P. Benítez, and J. C. Miñano, “Recent Trends in Concentrated Photovoltaics Concentrators’ Architecture,” *Journal of Photonics for Energy* 4, no. 1 (2014): 040995, <https://doi.org/10.1117/1.JPE.4.040995>.
 49. D. C. Miller, H. I. Khonkar, R. Herrero, et al., “An End of Service Life Assessment of PMMA Lenses from Veteran Concentrator Photovoltaic Systems,” *Solar Energy Materials and Solar Cells* 167 (2017): 7–21, <https://doi.org/10.1016/j.solmat.2017.03.031>.
 50. Eureka, “How PMMA Advances the Development of New Optical Materials?,” accessed December 14, 2025, <https://eureka.patsnap.com/report-how-pmma-advances-the-development-of-new-optical-materials>.
 51. K. Nishioka, K. Ikematsu, Y. Ota, and K. Araki, “Sandblasting Durability of Acrylic and Glass Fresnel Lenses for Concentrator Photovoltaic Modules,” *Solar Energy* 86, no. 10 (2012): 3021–3025, <https://doi.org/10.1016/j.solener.2012.07.009>.
 52. D. Sato, K. Tanino, and N. Yamada, “Thermal Characterization of a Silicone-on-Glass Micro Aspheric Lens Array for a Concentrator Photovoltaic Module,” *Solar Energy Materials and Solar Cells* 208 (2020): 110396, <https://doi.org/10.1016/j.solmat.2020.110396>.
 53. M. Frenzel, C. Mikolajczak, M. A. Reuter, and J. Gutzmer, “Quantifying the Relative Availability of High-Tech By-Product Metals – the Cases of Gallium, Germanium and Indium,” *Resources Policy* 52 (2017): 327–335, <https://doi.org/10.1016/J.RESOURPOL.2017.04.008>.
 54. M. Wiesenfarth, I. Anton, and A. W. Bett, “Challenges in the Design of Concentrator Photovoltaic (CPV) Modules to Achieve Highest Efficiencies,” *Applied Physics Reviews* 5, no. 4 (2018): 041601, <https://doi.org/10.1063/1.5046752>.
 55. C. Algora, “Very-High-Concentration Challenges of III-V Multijunction Solar Cells,” *Springer Series in Optical Sciences* 130 (2007): 89–111, https://doi.org/10.1007/978-3-540-68798-6_5/COVER.
 56. M. Baudrit, I. Rey-Stolle, et al., “Pending Issues in the Modeling of Concentrator Solar Cells,” in *Proceedings of the 19th European Photovoltaic Solar Energy Conference*, Paris, 2004.
 57. D. Il Lee, G. S. Oh, and S. W. Baek, “Development of a Heating Device Using CPV and Heat Pipe,” 2014, accessed August 25, 2025, <http://hdl.handle.net/2263/44627>.
 58. Z. Wang, D. Kortge, J. Zhu, et al., “Lightweight, Passive Radiative Cooling to Enhance Concentrating Photovoltaics,” *Joule* 4, no. 12 (2020): 2702–2717, <https://doi.org/10.1016/J.JOULE.2020.10.004>.
 59. K. Nishioka, Y. Ota, K. Tamura, and K. Araki, “Heat Reduction of Concentrator Photovoltaic Module Using High Radiation Coating,” *Surface & Coatings Technology* 215 (2013): 472–475, <https://doi.org/10.1016/J.SURFCOAT.2012.09.064>.
 60. Z. Zhou, Z. E. Wang, and P. Bermel, “Radiative Cooling for Low-Bandgap Photovoltaics under Concentrated Sunlight,” *Optics Express* 27, no. 8 (2019): A404–A418, <https://doi.org/10.1364/OE.27.00A404>.
 61. Z. Zhou, Y. Sun, X. Sun, M. A. Alam, P. Bermel, and X. Jin, “Radiative Cooling for Concentrating Photovoltaic Systems,” *Thermal Radiation Management for Energy Applications*, eds. M. M. Al-Jassim and P. Bermel (SPIE, 2017), 12, <https://doi.org/10.1117/12.2273916>.
 62. Y. Li, L. Zhang, W. Gao, et al., “Radiative Sky Cooling in Low-Medium Concentration Photovoltaic Systems,” *Applied Thermal Engineering* 221 (2023): 119860, <https://doi.org/10.1016/J.APPLTHERMALENG.2022.119860>.
 63. Z. Wang, *Radiative Passive Cooling for Concentrated Photovoltaics* (Purdue University, 2019), <https://doi.org/10.25394/PGS.11328182.V1>.
 64. H. Togun, A. Basem, M. J. Jweeg, et al., “Development and Innovation Using PCM in PV Cooling Systems: Passive and Active Approaches,” *Journal of Thermal Analysis and Calorimetry* 150, no. 14 (2025): 10725–10760, <https://doi.org/10.1007/s10973-025-14388-1>.
 65. K. Tabet Aoul, A. Hassan, A. H. Shah, and H. Riaz, “Energy Performance Comparison of Concentrated Photovoltaic – Phase Change Material Thermal (CPV-PCM/T) System with Flat Plate Collector (FPC),” *Solar Energy* 176 (2018): 453–464, <https://doi.org/10.1016/J.SOLENER.2018.10.039>.
 66. M. Emam and M. Ahmed, “Performance Analysis of a New Concentrator Photovoltaic System Integrated with Phase Change Material and Water Jacket,” *Solar Energy* 173 (2018): 1158–1172, <https://doi.org/10.1016/J.SOLENER.2018.08.069>.
 67. R. Rabie, M. Emam, A. Elwardany, S. Ookawara, and M. Ahmed, “Performance Evaluation of Concentrator Photovoltaic Systems Integrated with Combined Passive Cooling Techniques,” *Solar Energy* 228 (2021): 447–463, <https://doi.org/10.1016/J.SOLENER.2021.09.055>.
 68. X. Han, Y. Wang, and L. Zhu, “The Performance and Long-Term Stability of Silicon Concentrator Solar Cells Immersed in Dielectric Liquids,” *Energy Conversion and Management* 66 (2013): 189–198, <https://doi.org/10.1016/J.ENCONMAN.2012.10.009>.
 69. H. A. Hasan, J. S. Sherzaa, L. A. Abd, et al., “Study the Effect of Flow Water/Al₂O₃ Nanofluid Inside Mini-Channel for Cooling Concentrated Multi-Junction Solar Cell,” *Frontiers in Heat and Mass Transfer* 18 (2022): 1–8, <https://doi.org/10.5098/hmt.18.45>.
 70. Y. Sun, Y. Wang, L. Zhu, B. Yin, H. Xiang, and Q. Huang, “Direct Liquid-Immersion Cooling of Concentrator Silicon Solar Cells in a Linear Concentrating Photovoltaic Receiver,” *Energy* 65 (2014): 264–271, <https://doi.org/10.1016/J.ENERGY.2013.11.063>.
 71. X. Kang, Y. Wang, Q. Huang, Y. Cui, X. Shi, and Y. Sun, “Study on Direct-Contact Phase-Change Liquid Immersion Cooling Dense-Array Solar Cells under High Concentration Ratios,” *Energy Conversion and Management* 128 (2016): 95–103, <https://doi.org/10.1016/J.ENCONMAN.2016.09.073>.
 72. M. Sheikholeslami and Z. Khalili, Simulation for Impact of Nanofluid Spectral Splitter on Efficiency of Concentrated Solar Photovoltaic Thermal System,” *Sustainable Cities and Society* 101 (2024): 105139, <https://doi.org/10.1016/j.scs.2023.105139>.
 73. I. al Siyabi, K. Shanks, T. Mallick, and S. Sundaram, “Thermal Analysis of a Multi-Layer Microchannel Heat Sink for Cooling Concentrator Photovoltaic (CPV) Cells,” *AIP Conference Proceedings* 1881, no. 1 (2017): 070001, <https://doi.org/10.1063/1.5001434>.
 74. A. Radwan, M. Ahmed, and S. Ookawara, “Performance Enhancement of Concentrated Photovoltaic Systems Using a Microchannel Heat

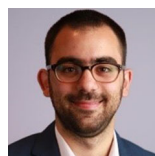
- Sink with Nanofluids," *Energy Conversion and Management* 119 (2016): 289–303, <https://doi.org/10.1016/J.ENCONMAN.2016.04.045>.
75. H. I. Elqady, E. M. Abo-Zahhad, A. Y. M. Ali, M. F. Elkady, A. H. El-Shazly, and A. Radwan, "Header Impact Assessment of Double-Layer Microchannel Heat Sink in the Computational Fluid Mechanics Simulation for CPV Thermal Management," *Energy Reports* 6 (2020): 55–60, <https://doi.org/10.1016/J.EGYR.2020.10.037>.
76. G. Laguna, J. Barrau, L. Fr  chette, et al., "Distributed and Self-Adaptive Microfluidic Cell Cooling for CPV Dense Array Receivers," *AIP Conference Proceedings* 1881, no. 1 (2017): 080006, <https://doi.org/10.1063/1.5001444>.
77. G. Rozsas, G. Bognar, G. Takacs, P. G. Szabo, and B. Plesz, "Process and Measurement of Electroplated Back-Contact Integrated Microchannel Cooling Devices for CPV Cells," in 2020 26th International Workshop on Thermal Investigations of ICs and Systems, THERMINIC (IEEE, 2020), <https://doi.org/10.1109/THERMINIC49743.2020.9420517>.
78. M. Awad, A. Radwan, O. Abdelrehim, M. Emam, A. N. Shmroukh, and M. Ahmed, "Performance Evaluation of Concentrator Photovoltaic Systems Integrated with a New Jet Impingement-Microchannel Heat Sink and Heat Spreader," *Solar Energy* 199 (2020): 852–863, <https://doi.org/10.1016/J.SOLENER.2020.02.078>.
79. E. M. Abo-Zahhad, S. Ookawara, A. Radwan, A. H. El-Shazly, and M. F. Elkady, "Thermal and Structure Analyses of High Concentrator Solar Cell under Confined Jet Impingement Cooling," *Energy Conversion and Management* 176 (2018): 39–54, <https://doi.org/10.1016/J.ENCONMAN.2018.09.005>.
80. A. Torbatinezhad, M. Rahimi, A. A. Ranjbar, and M. Gorzin, "Performance Evaluation of PV Cells in HCPV/T System by a Jet Impingement/Mini-Channel Cooling Scheme," *International Journal of Heat and Mass Transfer* 178 (2021): 121610, <https://doi.org/10.1016/J.IJHEATMASSTRANSFER.2021.121610>.
81. J. Tang, X. Li, R. Hu, Z. Mo, and M. Du, "A Novel Designed Manifold Ultrathin Micro Pin-Fin Channel for Thermal Management of High-Concentrator Photovoltaic System," *International Journal of Heat and Mass Transfer* 183 (2022): 122094, <https://doi.org/10.1016/J.IJHEATMASSTRANSFER.2021.122094>.
82. M. Sheikholeslami and A. M. Alinia, "Energy Management in a Concentrated Solar Photovoltaic Panel with a Thermoelectric Module and Nanomaterial-Filled Storage Tank," *Energy* 328 (2025): 136439, <https://doi.org/10.1016/j.energy.2025.136439>.
83. Y. Cai, W. W. Wang, C. W. Liu, W. T. Ding, D. Liu, and F. Y. Zhao, "Performance Evaluation of a Thermoelectric Ventilation System Driven by the Concentrated Photovoltaic Thermoelectric Generators for Green Building Operations," *Renewable Energy* 147 (2020): 1565–1583, <https://doi.org/10.1016/J.RENENE.2019.09.090>.
84. A. Abdo, T. Saito, S. Ookawara, A. Radwan, and M. Ahmed, "Experimental Study of the Performance of Concentrator Photovoltaic/Thermoelectric Generator System Integrated with a New 3D Printed Microchannel Heat Sink," *International Journal of Energy Research* 45, no. 5 (2021): 7741–7763, <https://doi.org/10.1002/ER.6359>.
85. O. Rejeb, S. Shittu, C. Ghenai, G. Li, X. Zhao, and M. Bettayeb, "Optimization and Performance Analysis of a Solar Concentrated Photovoltaic-Thermoelectric (CPV-TE) Hybrid System," *Renewable Energy* 152 (2020): 1342–1353, <https://doi.org/10.1016/J.RENENE.2020.02.007>.
86. Y. Zhou, Y. Chen, Q. Zhang, et al., "A Highly-Efficient Concentrated Perovskite Solar Cell-Thermoelectric Generator Tandem System," *Journal of Energy Chemistry* 59 (2021): 730–735, <https://doi.org/10.1016/j.jechem.2020.12.020>.
87. X. Han, Y. Sun, J. Huang, and J. Zheng, "Design and Analysis of a CPV/T Solar Receiver with Volumetric Absorption Combined Spectral Splitter," *International Journal of Energy Research* 44, no. 6 (2020): 4837–4850, <https://doi.org/10.1002/ER.5277>.
88. G. Wang, Y. Yao, J. Lin, Z. Chen, and P. Hu, "Design and Thermodynamic Analysis of a Novel Solar CPV and Thermal Combined System Utilizing Spectral Beam Splitter," *Renewable Energy* 155 (2020): 1091–1102, <https://doi.org/10.1016/J.RENENE.2020.04.024>.
89. X. Han, L. Tu, and Y. Sun, "A Spectrally Splitting Concentrating PV/T System Using Combined Absorption Optical Filter and Linear Fresnel Reflector Concentrator," *Solar Energy* 223 (2021): 168–181, <https://doi.org/10.1016/J.SOLENER.2021.05.039>.
90. A. Mohammadnia and B. M. Ziapour, "Investigation Effect of a Spectral Beam Splitter on Performance of a Hybrid CPV/Stirling/TEG Solar Power System," *Applied Thermal Engineering* 180 (2020): 115799, <https://doi.org/10.1016/J.APPLTHERMALENG.2020.115799>.
91. Y. Zhu, B. Ma, Z. Zeng, and D. Jing, "Multi-Functional Solar Collector Tube with Significantly Enhanced Optical Performance for Improved CPV/T System Output," *SSRN Electronic Journal* (2021), <https://doi.org/10.2139/SSRN.3994681>.
92. T. Kinoshita, K. Nonomura, N. J. Jeon, et al., "Spectral Splitting Photovoltaics Using Perovskite and Wideband Dye-Sensitized Solar Cells," *Nature Communications*, 6, no. 1 (2015), 8834, <https://doi.org/10.1038/ncomms9834>.
93. M. Wiesenfarth, J. Went, A. B  sch, et al., "CPV-T Mirror Dish System Combined with Water Desalination Systems," *AIP Conference Proceedings* 1766, no. 1 (2016): 020008, <https://doi.org/10.1063/1.4962076>.
94. A. Khouya, "4E Assessment of a Hybrid RO/MED-TVC Desalination Plant Powered by CPV/T Systems," *Energy Conversion and Management* 277 (2023): 116666, <https://doi.org/10.1016/J.ENCONMAN.2023.116666>.
95. S. Liang, H. Zheng, Z. Zhao, X. Ma, and K. C. Ng, "Investigation on an Underwater Solar Concentrating Photovoltaic-Membrane Distillation (CPV-MD) Integrated System," *Desalination* 546 (2023): 116193, <https://doi.org/10.1016/J.DESAL.2022.116193>.
96. B. Anand and T. Srinivas, "Cogeneration of Power and Desalination Using Concentrated Photovoltaic/Thermal Humidification and Dehumidification System," *Advances in Energy Research*, 1 (2020): 125–137, https://doi.org/10.1007/978-981-15-2666-4_13.
97. I. Albaik, Y. A. Alamri, M. B. Elsheniti, R. Al-Dadah, S. Mahmoud, and M. A. Ismail, "Assessment of a Novel Multi-Generation Solar CPV/T System Combining Adsorption and Organic Rankine Cycle Subsystems," *Solar Energy* 236 (2022): 455–472, <https://doi.org/10.1016/J.SOLENER.2022.03.024>.
98. C. Lai, H. Zhang, and Z. Hu, "Performance Analyses on a Concentrated Photovoltaics Driven Direct Contact Membrane Distillation Coupled System," *Desalination* 531 (2022): 115695, <https://doi.org/10.1016/J.DESAL.2022.115695>.
99. S. Sharma, N. Sellami, A. A. Tahir, T. K. Mallick, and R. Bhakar, "Performance Improvement of a CPV System: Experimental Investigation into Passive Cooling with Phase Change Materials," *Energies*, 14, no. 12 (2021): 3550, <https://doi.org/10.3390/EN14123550>.
100. M. I. Hussain and J.-T. Kim, "Energy and Economic Potential of a Concentrated Photovoltaic/Thermal (CPV/T) System for Buildings in South Korea," *Journal of Asian Architecture and Building Engineering* 18, no. 2 (2019): 139–144, <https://doi.org/10.1080/13467581.2019.1606718>.
101. L. Zhu, J. Zhang, Q. Li, et al., "Comprehensive Analysis of Heat Transfer of Double-Skin Facades Integrated High Concentration Photovoltaic (CPV-DSF)," *Renewable Energy* 161 (2020): 635–649, <https://doi.org/10.1016/J.RENENE.2020.07.045>.
102. S. Liang, H. Zheng, S. Liu, and X. Ma, "Optical Design and Validation of a Solar Concentrating Photovoltaic-Thermal (CPV-T) Module for Building Louvers," *Energy* 239 (2022): 122256, <https://doi.org/10.1016/J.JENERGY.2021.122256>.

103. D. L. Talavera, P. Pérez-Higueras, J. A. Ruiz-Arias, and E. F. Fernández, "Levelised Cost of Electricity in High Concentrated Photovoltaic Grid Connected Systems: Spatial Analysis of Spain," *Applied Energy* 151 (2015): 49–59, <https://doi.org/10.1016/J.APENERGY.2015.04.072>.
104. E. F. Fernandez, N. Seoane, F. Almonacid, and A. J. Garcia-Loureiro, "Vertical-Tunnel-Junction (VTJ) Solar Cell for Ultra-High Light Concentrations (>2000 Suns)," *IEEE Electron Device Letters* 40, no. 1 (2018): 1–47, <https://doi.org/10.1109/LED.2018.2880240>.
105. L. Micheli, E. F. Fernandez, F. Almonacid, K. S. Reddy, and T. K. Mallick, "Enhancing Ultra-High CPV Passive Cooling Using Least-Material Finned Heat Sinks," *AIP Conference Proceedings* 1679, no. 1 (2015): 130003, <https://doi.org/10.1063/1.4931563>.
106. F. Almonacid, E. F. Fernandez, A. Mellit, and S. Kalogirou, "Review of Techniques Based on Artificial Neural Networks for the Electrical Characterization of Concentrator Photovoltaic Technology," *Renewable and Sustainable Energy Reviews* 75 (2017): 938–953, <https://doi.org/10.1016/j.rser.2016.11.075>.
107. F. Masood, P. Nallagownden, I. Elamvazuthi, J. Akhter, and M. A. Alam, "A Review of Artificial Intelligence-Based Techniques to Estimate Atmospheric Parameters Influencing the Performance of Concentrating Photovoltaic/Thermal Systems," *International Conference on Artificial Intelligence for Smart Community. Lecture Notes in Electrical Engineering* 758 (2022): 365–372, https://doi.org/10.1007/978-981-16-2183-3_35.
108. M. K. Kolamroudi, O. H. Jaiyeoba, M. Ilkan, and B. Safaei, "A Comprehensive Review on the Artificial Intelligence for the Development of Thermal Concentrating Photovoltaic Systems," *Solar Energy* 301 (2025): 113937, <https://doi.org/10.1016/j.solener.2025.113937>.
109. S. C. Nwokolo, et al., Machine Learning and Physics-Based Model Hybridization to Assess the Impact of Climate Change on Single- and Dual-Axis Tracking Solar-Concentrated Photovoltaic Systems," *Physics and Chemistry of the Earth, Parts A/B/C* 138 (2025): 103881, <https://doi.org/10.1016/j.pce.2025.103881>.
110. K. Moradi, F. Jafari, F. Moghaddam, and Q. Shafiee, "Enhancing the Performance of Concentrated Photovoltaic-Thermal Systems Using Solar Tracking Mirrors," *Renewable Energy* 248 (2025): 122978, <https://doi.org/10.1016/j.renene.2025.122978>.
111. V. V. R. Raju and P. Juge, "IoT-Based Concentrated Photovoltaic Solar System," in *Blockchain Technology for IoT and Wireless Communications* (CRC Press, 2023).
112. J. Canada-Bago, J.-A. Fernandez-Prieto, M.-A. Gadeo-Martos, and P. Perez-Higueras, "Knowledge-Based Sensors for Controlling A High-Concentration Photovoltaic Tracker," *Sensors* 20, no. 5 (2020): 1315, <https://doi.org/10.3390/s20051315>.

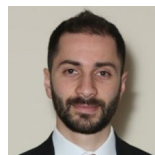
Biographies



Prof. Abdul Ghani Olabi is the Director of Sustainable Energy and Power Systems Research Centre at the University of Sharjah "UoS." Previously, he was Head of Department, Director of research institute, and many other academic and industrial position. He has supervised postgraduate research students (38 Ph.D.) to successful completion. He has patented 2 innovative projects on PEM Fuel Cell. He is the Subject Editor of the Elsevier Energy Journal, EiC of the Encyclopedia of Smart Materials (Elsevier), Editor of the Reference Module of Materials Science and Engineering (Elsevier), EiC of Renewable Energy section of Energies and board member of few other journals. Professor Olabi published more than 350 manuscripts in ISI indexed journals.



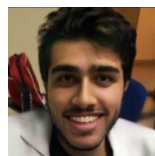
Adnan Alashkar is currently working as a post-doctoral research fellow in the Research Institute of Sciences and Engineering, University of Sharjah. He received his Ph.D. in Materials Science and Engineering from the American University of Sharjah in 2025. His research interests include renewable energy systems, third-generation photovoltaics, sustainable and additive manufacturing, and mechanical energy storage systems, with a particular focus on compressed air energy storage.



Montaser Mahmoud is a Post-Doctoral Researcher at the University of Sharjah (UoS) in the Sustainable Energy and Power Systems Research Centre. He got his Ph.D. in Mechanical Engineering from City University of London, United Kingdom, in 2022, following a B.Sc. and M.Sc. in Mechanical Engineering from the Lebanese International University in 2015 and 2017, respectively. His research interests include renewable energy, energy storage systems, energy management, and hybrid energy systems. Dr. Montaser Mahmoud possesses advanced skills in engineering software, including SolidWorks, MATLAB, ANSYS, EES, and AutoCAD, and has contributed significantly to the energy research field through numerous journal articles and book chapters.



Ayman Mdallal is a civil engineering researcher specialized in renewable energy and low-carbon concrete. He earned his B.Sc. in Civil Engineering from Jordan University of Science and Technology in 2019, followed by an M.Sc. in Renewable Energy/Mechanical Engineering in 2021 from the University of Jordan. From 2022 to 2024, he worked as a Graduate Research Assistant at the University of Sharjah, contributing to research in renewable energy and sustainability. He is currently pursuing a Ph.D. in Civil Engineering at McMaster University, focusing on the development and performance of low-carbon concrete and sustainable cementitious materials.



Malek Kamal Rabaia is a Graduate Research Assistant under the Research Institute of Sciences and Engineering at the University of Sharjah, United Arab Emirates, where he got a B.Sc. in Sustainable and Renewable Energy Engineering and an M.Sc. in Engineering Management. His Research interests include Sustainable Supply Chain Management (also called Circular Economy Design), Solar PV, Renewable Energy, Digital Twins, and Automation, mainly employing Mathematical Modeling & Optimization (e.g., LP, MIP, MINLP, etc.), and Data Management for Advanced Machine Learning.



Abdul Hai Alami is currently a professor and Chairman of the Sustainable and Renewable Energy Engineering Program at University of Sharjah. Professor Alami has received his Ph.D. from Queen's University in Kingston, Ontario, Canada, in 2006. Prof. Alami has been among world's top 2% scientists according to Stanford University Scientists Ranking since 2020. The current areas of interest of Prof. Alami is the synthesis and analysis of materials used in third generation photovoltaic solar cells, solar thermal energy utilization and augmentation (selective solar absorbers, evaporative cooling of PV modules), and novel ways of mechanical energy storage (CAES, CGES, and buoyancy force). Prof. Alami has authored around 200 publications in highly reputed journals, as well as books on energy storage technologies and solar photovoltaic manufacturing.



Dr. Mahmoud Elgendi is an Associate Professor of Thermo-Fluid Engineering at United Arab Emirates University (UAEU). He received his Ph.D. from Nagoya University, Japan, and his M.Sc. from Minia University, Egypt. He completed a postdoctoral fellowship at Ulsan University, South Korea, in collaboration with Hyundai Heavy Industry, and served as a visiting scholar at the University of Illinois Urbana-Champaign, USA. His research interests

include thermo-fluid systems, turbomachinery, computational fluid dynamics, energy, and heat transfer.



Prof. Mohammad Ali Abdelkareem got his Ph.D. from Japan in 2008. Right now, he is professor in sustainable and renewable energy engineering department, university of Sharjah, UAE. He is working on the development of the different renewable energy resources, thermofluids, and electrochemical devices that can be used in wastewater treatment and water desalination. Moreover, he is working on the development of the electrodes for the different electrochemical energy conversion/storage devices, such as direct methanol fuel cells, direct urea fuel cells, microbial fuel cells, and supercapacitors. Professor Mohammad published more than 200 manuscripts in ISI indexed journals.